

Interpolation on a Geometric Comb

Lagrange filters, Jackson–Newton series, q -analogues, and the Fabius–Rvachev bridge

Independent research report for Vladimir Reshetnikov’s ProveIt project

30 August 2026

Abstract

Let

$$\mathcal{C}_{a,q} = \{a, aq, aq^2, \dots\}, \quad a \neq 0, \quad 0 < q < 1,$$

be a one-sided geometric comb accumulating at the origin. Polynomial interpolation on its first $n + 1$ teeth is governed exactly by finite q -products. This report develops that observation into a unified theory. The geometric Vandermonde matrix factors into a Gaussian Pascal matrix and a diagonal q -factorial matrix; the nodal polynomial is a q -Pochhammer symbol; barycentric weights and arbitrary-target cardinal values have closed q -gamma forms; and the Newton divided differences are precisely normalized Jackson derivatives,

$$f[a, aq, \dots, aq^k] = \frac{D_q^k f(a)}{[k]_q!}.$$

The resulting extrapolation operator at the accumulation point is the finite dilation polynomial

$$\mathcal{G}_{n,q} = \frac{1}{(q; q)_n} \prod_{j=1}^n (E_q - q^j I), \quad (E_q f)(x) = f(qx).$$

It is therefore diagonal on Mellin monomials. For every complex s ,

$$\mathcal{G}_{n,q}(x^s)|_{x=a} = a^s q^{ns} \frac{(q^{1-s}; q)_n}{(q; q)_n}.$$

This formula extends the familiar integer moment annihilation to fractional powers and logarithmic powers and makes the relation with Richardson–Romberg extrapolation transparent. The exact variation of the zero-evaluation row is

$$\sum_{k=0}^n |\lambda_{n,k}^{(q)}| = \frac{(-q; q)_n}{(q; q)_n},$$

so, for fixed $q < 1$, extrapolation to the accumulation point is a regular summability method even though ordinary interpolation away from the comb can be violently unstable.

The second half constructs the geometric-uniform family

$$X_q = (1 - q) \sum_{j \geq 0} q^j U_j, \quad U_j \sim \text{Unif}[0, 1],$$

with distribution function F_q and centered density u_q . At $q = 1/2$, F_q is the bounded Fabius function and u_q is Rvachev’s up-function. For $0 < q \leq 1/2$ we prove the exact comb-moment identity

$$F_q(q^n) = \frac{q^{n(n+1)/2}}{(1 - q)^n n!} \mathbb{E} X_q^n.$$

At the dyadic point this converts the Fabius interpolation table into an explicit Gaussian/Mersenne transform of the moment sequence. If A_n is the Newton coefficient multiplying $\prod_{r < n} (x - 2^{-r})$, then

$$A_n = \frac{1}{(1/2; 1/2)_n} \sum_{j=0}^n (-1)^j \begin{bmatrix} n \\ j \end{bmatrix}_2 \frac{\mathbb{E} X^j}{j!} = 2^{n(n+1)/2} \sum_{j=0}^n \frac{(-1)^j \mathbb{E} X^j}{j! \mathfrak{M}_j \mathfrak{M}_{n-j}},$$

where $\mathfrak{M}_r = \prod_{m=1}^r (2^m - 1)$. The extrapolants at zero converge superexponentially, with a proved $2^{-n^2/4 + O(n)}$ bound. In sharp contrast, flatness implies that $\limsup |A_n|^{1/n} = \infty$, and hence the Fabius Newton interpolants diverge at every nonzero point other than the sampled powers of $1/2$. Thus their exact convergence set is the comb together with its accumulation point.

Finally, the Fourier product

$$\widehat{u}_q(\xi) = \prod_{j \geq 0} \operatorname{sinc}_{\pi}(2(1-q)q^j \xi)$$

is fed back through the same geometric filter. Away from its zeros this gives a rapidly convergent reciprocal-product expansion for $1/\widehat{u}_q$, with an explicit factorial–Gaussian error bound. The report also relates the multiplicative Gaussian filter to dyadic Thue–Morse spline formulas, to exact finite synthesis by shifted up-functions, and to a collection of concrete formalization and research problems. An accompanying Python program verifies the finite identities exactly, computes Fabius moments through order 80, and reproduces all numerical figures.

Provenance and novelty discipline

The starting monograph already contains the finite geometric Newton formula, the reversed-Gaussian zero-evaluation row, its natural-number moment law, and the associated formal power-series filter. The adjacent repository report on the Lagrange–Legendre–Rvachev loop treats arbitrary-node Lagrange cardinals and finite up-atom synthesis. The present report reorganizes those ingredients around the geometric comb and adds, with proofs, the Gaussian-Pascal matrix factorization, arbitrary geometric targets, the exact row-variation formula, the full complex Mellin symbol, logarithmic-power responses, analytic infinite Jackson–Newton series, a regular-summability theorem, the superflat nonconvergence theorem, the geometric-uniform comb-moment formula, the exact Fabius Gaussian/Mersenne transforms, the quadratic endpoint bound, and the reciprocal Rvachev product filter. Claims labeled as conjectures are not used in any proof. “New” means new relative to the inspected live repository corpus, not a claim of worldwide priority.

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1 Orientation and the four equivalent pictures

1.1 The geometric comb

Fix $a \in \mathbb{C} \setminus \{0\}$ and a base $q \in \mathbb{C}$ such that $q^i \neq q^j$ for the powers under consideration. The finite geometric comb of order n is

$$\mathcal{C}_{a,q}^{(n)} = \{x_j = aq^j : 0 \leq j \leq n\}.$$

Most analytic statements below assume $0 < q < 1$. Algebraic formulas remain valid for arbitrary nonzero q until two nodes collide. An affine comb $b + aq^j$ is reduced to the present one by the change of variable $x \mapsto x - b$; keeping $b = 0$ exposes the multiplicative structure and the accumulation point.

Four descriptions of the same interpolation problem will be used repeatedly.

1. In the original variable x , the Newton factors are $x - a, x - aq, x - aq^2, \dots$ and hence are finite q -Pochhammer products.
2. In the logarithmic variable $x = ae^{-t}$ with $q = e^{-h}$, the nodes are the arithmetic grid $t = 0, h, 2h, \dots$, while a degree- n polynomial in x becomes an exponential polynomial in t spanned by $1, e^{-t}, \dots, e^{-nt}$.
3. On sequences $f(aq^k)$, interpolation at $x = 0$ is a finite polynomial in the dilation operator $E_q : f(x) \mapsto f(qx)$.
4. Mellin monomials x^s diagonalize dilations. The interpolation filter consequently has an exact scalar symbol expressed by a q -Pochhammer quotient.

The first two pictures explain why geometric interpolation was already natural to Stirling and Schellbach; the third is the language of Richardson and Romberg extrapolation; the fourth is the most efficient bridge to fractional powers, endpoint asymptotics, and the Fabius/Rvachev system. Schoenberg's historical and numerical account [13] explicitly connects these traditions.

1.2 Basic notation

For $n \in \mathbb{N}_0$,

$$(z; q)_n = \prod_{r=0}^{n-1} (1 - zq^r), \quad (z; q)_\infty = \prod_{r=0}^{\infty} (1 - zq^r)$$

when the infinite product converges. The q -integer, q -factorial, and Gaussian coefficient are

$$[m]_q = \frac{1 - q^m}{1 - q}, \quad [m]_q! = \prod_{r=1}^m [r]_q = \frac{(q; q)_m}{(1 - q)^m}, \quad \begin{bmatrix} n \\ k \end{bmatrix}_q = \frac{(q; q)_n}{(q; q)_k (q; q)_{n-k}}.$$

The Gaussian coefficient is set to zero when $k \notin \{0, \dots, n\}$. We use Euler's finite q -binomial theorem in the form

$$(z; q)_n = \sum_{r=0}^n (-1)^r q^{\binom{r}{2}} \begin{bmatrix} n \\ r \end{bmatrix}_q z^r. \quad (1)$$

Standard references for this notation and its analytic continuations are Gasper–Rahman [5], Kac–Cheung [8], and the DLMF [3].

1.3 A map of the proved results

The algebraic core consists of three triangular transforms:

$$\text{monomials} \longleftrightarrow \text{geometric Newton products} \longleftrightarrow \text{sample values on } \mathcal{C}_{a,q}^{(n)}.$$

All three transforms are Gaussian. The analytic core then separates into two opposite facts.

- At the accumulation point, the Lagrange rows have uniformly bounded total variation, so they define a regular summability method for every continuous germ.
- Away from the comb, data that are nonzero but flat at the accumulation point force superexponential Newton coefficients and divergence.

The Fabius function realizes both extremes simultaneously: exact dyadic values make the Gaussian transforms arithmetic, while endpoint flatness makes the interpolants converge extraordinarily fast at 0 and diverge extraordinarily fast almost everywhere else.

2 Algebra of the geometric Vandermonde system

2.1 Nodal polynomial and Vandermonde determinant

Proposition 2.1 (The nodal polynomial is a shifted factorial). *For the nodes $x_j = aq^j$, $0 \leq j \leq n$, put*

$$\Omega_{n+1}(x) = \prod_{j=0}^n (x - aq^j).$$

Then

$$\Omega_{n+1}(x) = x^{n+1} (a/x; q)_{n+1}, \quad (2)$$

$$= (-a)^{n+1} q^{\binom{n+1}{2}} (x/a; q^{-1})_{n+1}. \quad (3)$$

The geometric Vandermonde determinant is

$$\prod_{0 \leq i < j \leq n} (aq^j - aq^i) = (-1)^{\binom{n+1}{2}} a^{\binom{n+1}{2}} q^{\binom{n+1}{3}} \prod_{d=1}^n (1 - q^d)^{n+1-d}. \quad (4)$$

Proof. The first identity is obtained by factoring x from every factor. For the second, write $x - aq^j = -aq^j(1 - xq^{-j}/a)$ and multiply the powers of q . For the determinant, each pair contributes $q^j - q^i = -q^i(1 - q^{j-i})$. There are $n+1-d$ pairs with difference d , and

$$\sum_{0 \leq i < j \leq n} i = \sum_{i=0}^{n-1} i(n-i) = \binom{n+1}{3}.$$

Multiplication gives (4). □

Remark 2.2. Formula (4) is already a conditioning warning. For fixed n , it vanishes to order $\binom{n+1}{2}$ as $q \rightarrow 1$; for fixed $q < 1$, it contains the superexponentially small factor $q^{\binom{n+1}{3}}$. These two mechanisms correspond, respectively, to coalescence of all nodes at a and accumulation of the late nodes at 0.

2.2 The Gaussian Pascal factorization

Define the geometric Newton products

$$\Phi_0(x) = 1, \quad \Phi_k(x) = \prod_{r=0}^{k-1} (x - aq^r) = (-a)^k q^{\binom{k}{2}} (x/a; q^{-1})_k. \quad (5)$$

Theorem 2.3 (Evaluation matrix and Gaussian Pascal matrix). *For $0 \leq k, j \leq n$,*

$$\Phi_k(aq^j) = \begin{cases} (-a)^k q^{\binom{k}{2}} (q; q)_k [j]_k, & j \geq k, \\ 0, & j < k. \end{cases} \quad (6)$$

Consequently the lower-triangular evaluation matrix $V_{j,k}^N = \Phi_k(aq^j)$ factors as

$$V^N = P_q D_{a,q}, \quad (P_q)_{j,k} = \begin{bmatrix} j \\ k \end{bmatrix}_q, \quad (D_{a,q})_{k,k} = (-a)^k q^{\binom{k}{2}} (q; q)_k. \quad (7)$$

The inverse Gaussian Pascal matrix is

$$(P_q^{-1})_{j,k} = (-1)^{j-k} q^{\binom{j-k}{2}} \begin{bmatrix} j \\ k \end{bmatrix}_q \quad (j \geq k). \quad (8)$$

Proof. If $j < k$, the product $\Phi_k(aq^j)$ contains its zero factor. If $j \geq k$,

$$\begin{aligned} \Phi_k(aq^j) &= \prod_{r=0}^{k-1} (aq^j - aq^r) \\ &= (-a)^k q^{\binom{k}{2}} \prod_{r=0}^{k-1} (1 - q^{j-r}) \\ &= (-a)^k q^{\binom{k}{2}} \frac{(q; q)_j}{(q; q)_{j-k}} \\ &= (-a)^k q^{\binom{k}{2}} (q; q)_k \begin{bmatrix} j \\ k \end{bmatrix}_q. \end{aligned}$$

This proves the factorization. To verify the inverse, multiply the proposed matrices. After the standard Gaussian convolution $\begin{bmatrix} j \\ r \end{bmatrix}_q \begin{bmatrix} r \\ k \end{bmatrix}_q = \begin{bmatrix} j \\ k \end{bmatrix}_q \begin{bmatrix} j-k \\ r-k \end{bmatrix}_q$, the off-diagonal entry is a scalar multiple of

$$\sum_{s=0}^{j-k} (-1)^s q^{\binom{s}{2}} \begin{bmatrix} j-k \\ s \end{bmatrix}_q = (1; q)_{j-k} = 0.$$

The diagonal entry is 1. □

Remark 2.4 (Why Gaussian coefficients are unavoidable). The matrix q^{jm} of monomial values on the comb is simultaneously a Vandermonde matrix and a finite Laplace-transform matrix. Factoring it through Newton products produces P_q . Thus Gaussian coefficients do not enter by analogy or decoration: they are the exact incidence coefficients of the geometric evaluation map.

2.3 Two inverse basis changes

Theorem 2.5 (Monomial–Newton connection coefficients). *For every $k, m \geq 0$,*

$$\Phi_k(x) = \sum_{j=0}^k (-1)^{k-j} a^{k-j} q^{\binom{k-j}{2}} \begin{bmatrix} k \\ j \end{bmatrix}_q x^j, \quad (9)$$

$$x^m = \sum_{k=0}^m \begin{bmatrix} m \\ k \end{bmatrix}_q a^{m-k} \Phi_k(x). \quad (10)$$

Proof. Expand $x^k(a/x; q)_k$ by (1) and put $j = k - r$ to get (9). For the inverse, either apply (8), or note that the coefficient of Φ_k in the Newton interpolant of x^m is its divided difference on $a, aq, \dots, aq^k$. The direct calculation in Theorem 5.1 below gives $a^{m-k} \begin{bmatrix} m \\ k \end{bmatrix}_q$. □

Corollary 2.6 (Newton coefficients of a formal power series). *If $f(x) = \sum_{m \geq 0} c_m x^m$ is a formal power series, then its formal geometric Newton coefficients are*

$$A_k = \sum_{r \geq 0} c_{k+r} \begin{bmatrix} k+r \\ k \end{bmatrix}_q a^r. \quad (11)$$

Whenever rearrangement is justified, $f(x) = \sum_{k \geq 0} A_k \Phi_k(x)$.

Proof. Insert (10) into each monomial and collect the coefficient of Φ_k . □

3 Lagrange cardinal functions on the comb

3.1 Barycentric weights

Let $\ell_{n,j}$ denote the cardinal polynomial satisfying $\ell_{n,j}(aq^r) = \delta_{jr}$ for $0 \leq r \leq n$.

Theorem 3.1 (Closed barycentric weights). *The literal barycentric weights $\beta_{n,j} = 1/\Omega'_{n+1}(aq^j)$ are*

$$\beta_{n,j} = \frac{(-1)^j q^{-jn + \binom{j+1}{2}}}{a^n (q; q)_j (q; q)_{n-j}}. \quad (12)$$

Their consecutive ratio is

$$\frac{\beta_{n,j+1}}{\beta_{n,j}} = -q^{j+1-n} \frac{1 - q^{n-j}}{1 - q^{j+1}}. \quad (13)$$

For x not equal to a node,

$$(\mathcal{I}f)(x) = \frac{\sum_{j=0}^n \frac{\beta_{n,j} f(aq^j)}{x - aq^j}}{\sum_{j=0}^n \frac{\beta_{n,j}}{x - aq^j}}. \quad (14)$$

Proof. Split the derivative product into indices below and above j :

$$\begin{aligned} \prod_{\substack{0 \leq r \leq n \\ r \neq j}} (aq^j - aq^r) &= a^n \left[\prod_{r < j} -q^r (1 - q^{j-r}) \right] \left[\prod_{r > j} q^j (1 - q^{r-j}) \right] \\ &= a^n (-1)^j q^{jn - \binom{j+1}{2}} (q; q)_j (q; q)_{n-j}. \end{aligned}$$

Taking the reciprocal gives (12); division of two successive terms gives (13). The barycentric formula is the usual identity $\ell_{n,j}(x) = \beta_{n,j}/(x - aq^j) / \sum_r \beta_{n,r}/(x - aq^r)$. $\square$

Algorithm 3.2 (Stable construction of the weights). The common scale of barycentric weights cancels in (14). Set an arbitrary nonzero $\tilde{\beta}_{n,0}$ and generate the rest by (13), renormalizing the whole row whenever magnitudes leave a safe floating-point range. This avoids separately forming $q^{-jn + \binom{j+1}{2}}$ and the small products $(q; q)_j$.

3.2 A target on the same multiplicative line

A target $x = aq^\alpha$, where α need not be integral, preserves all q -structure.

Theorem 3.3 (Arbitrary geometric target). *Choose a branch of q^α . If q^α is not one of $1, q, \dots, q^n$, then*

$$\ell_{n,j}(aq^\alpha) = (-1)^{n-j} q^{\binom{n-j+1}{2}} \frac{(q^{\alpha-n}; q)_{n+1}}{(1 - q^{\alpha-j})(q; q)_j (q; q)_{n-j}}. \quad (15)$$

At $\alpha = j$ the singularity is removable and the value is 1; at every other integral node it is 0. For $0 < q < 1$ and $\Re \alpha \rightarrow +\infty$, the weights tend to the zero-target row

$$\lambda_{n,j}^{(q)} = \ell_{n,j}(0) = \frac{(-1)^{n-j} q^{\binom{n-j+1}{2}}}{(q; q)_j (q; q)_{n-j}}. \quad (16)$$

Proof. Use $\ell_{n,j}(x) = \Omega_{n+1}(x)/((x - aq^j)\Omega'_{n+1}(aq^j))$, Proposition 2.1, and (12). The product at the target is

$$\prod_{r=0}^n (q^\alpha - q^r) = (-1)^{n+1} q^{\binom{n+1}{2}} (q^{\alpha-n}; q)_{n+1}.$$

Extracting $-q^j$ from $q^\alpha - q^j$ leaves exactly (15). The limiting assertions follow directly from the product. $\square$

Remark 3.4 (q -gamma form). Since

$$(q^{\alpha-n}; q)_{n+1} = (1-q)^{n+1} \frac{\Gamma_q(\alpha+1)}{\Gamma_q(\alpha-n)},$$

formula (15) is meromorphic in α and can be rewritten entirely with q -gamma functions and the q -number $[\alpha - j]_q$. This is the natural analytic continuation of the cardinal row, not merely a formal replacement of factorials by q -factorials.

3.3 The zero row as a dilation polynomial

Theorem 3.5 (Reversed Gaussian characteristic identity). *For every indeterminate z ,*

$$\sum_{k=0}^n \lambda_{n,k}^{(q)} z^k = \frac{1}{(q; q)_n} \prod_{j=1}^n (z - q^j). \quad (17)$$

Consequently, with $(E_q f)(x) = f(qx)$,

$$(\mathcal{G}_{n,q} f)(x) := \sum_{k=0}^n \lambda_{n,k}^{(q)} f(q^k x) = \frac{1}{(q; q)_n} \prod_{j=1}^n (E_q - q^j I) f(x). \quad (18)$$

The operator reproduces constants and annihilates $x, x^2, \dots, x^n$.

Proof. Expand the product after reversing the finite q -binomial theorem. More explicitly, the coefficient of z^k in $\prod_{j=1}^n (z - q^j)$ is

$$(-1)^{n-k} q^{\binom{n-k+1}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_q,$$

and division by $(q; q)_n$ gives (16). Substitution of E_q for z proves (18). On x^d , E_q has eigenvalue q^d , so one factor vanishes when $1 \leq d \leq n$; the value on a constant is 1. $\square$

Remark 3.6 (Denominator-free form). Multiplying (17) by $(q; q)_n$ produces an identity in $\mathbb{Z}[q, z]$. It remains meaningful at roots of unity even when the normalized Lagrange row is undefined because nodes collide. This denominator-free residual is the right starting point for confluent and cyclotomic versions of the theory.

3.4 Exact total variation and conditioning at the accumulation point

Theorem 3.7 (Exact row norm). *For $0 < q < 1$,*

$$\sum_{k=0}^n |\lambda_{n,k}^{(q)}| = \frac{(-q; q)_n}{(q; q)_n} = \prod_{j=1}^n \frac{1+q^j}{1-q^j}. \quad (19)$$

In particular the row norms increase to the finite limit

$$K(q) = \frac{(-q; q)_\infty}{(q; q)_\infty}. \quad (20)$$

For fixed n and $q \rightarrow 1^-$,

$$\sum_{k=0}^n |\lambda_{n,k}^{(q)}| \sim \frac{2^n}{n!(1-q)^n}. \quad (21)$$

If $q = e^{-h}$ and $h \rightarrow 0^+$, then

$$K(e^{-h}) \sim \frac{\sqrt{h}}{2\sqrt{\pi}} \exp\left(\frac{\pi^2}{4h}\right). \quad (22)$$

Proof. Because the signs alternate,

$$\begin{aligned} \sum_{k=0}^n |\lambda_{n,k}^{(q)}| &= \frac{1}{(q; q)_n} \sum_{r=0}^n q^{\binom{r+1}{2}} \begin{bmatrix} n \\ r \end{bmatrix}_q \\ &= \frac{(-q; q)_n}{(q; q)_n}, \end{aligned}$$

where $r = n - k$ and the last step is the finite q -binomial theorem applied to $\prod_{j=1}^n (1 + q^j)$. The infinite product converges for $q < 1$. For fixed n , use $1 - q^j \sim j(1 - q)$ and $1 + q^j \sim 2$. Finally,

$$\frac{(-q; q)_\infty}{(q; q)_\infty} = \frac{(q^2; q^2)_\infty}{(q; q)_\infty^2}.$$

The standard Dedekind-eta asymptotic $(e^{-h}; e^{-h})_\infty \sim \sqrt{2\pi/h} \exp(-\pi^2/(6h) + h/24)$ gives (22) after substitution at h and $2h$. $\square$

Remark 3.8. The finite condition number blows up when q approaches 1, but for every fixed $q < 1$ the entire sequence of extrapolation rows has bounded variation. This distinction is essential. The accumulation point is stable as a summability target at fixed ratio, even though coalescing ratios are extremely ill-conditioned.

4 Mellin symbols and exact residuals

4.1 Complex powers

The integer moment law in the source monograph is one specialization of a stronger identity. Since dilations act diagonally on every chosen branch of x^s , the full spectral response is available at once.

Theorem 4.1 (Exact Mellin symbol). *Let $s \in \mathbb{C}$, choose a branch of a^s and q^s , and set*

$$M_{n,q}(s) = \sum_{k=0}^n \lambda_{n,k}^{(q)} q^{ks}. \quad (23)$$

Then

$$M_{n,q}(s) = \frac{\prod_{j=1}^n (q^s - q^j)}{(q; q)_n} \quad (24)$$

$$= q^{ns} \frac{(q^{1-s}; q)_n}{(q; q)_n} \quad (25)$$

$$= (-1)^n q^{\binom{n+1}{2}} \frac{(q^{s-n}; q)_n}{(q; q)_n}. \quad (26)$$

Equivalently,

$$(\mathcal{G}_{n,q} x^s)(a) = a^s M_{n,q}(s). \quad (27)$$

The zeros at $s = 1, \dots, n$ are simple modulo the imaginary period $2\pi i / \log q$.

Proof. Evaluate (17) at $z = q^s$. Factoring q^s from each factor gives (25); factoring $-q^j$ gives (26). $\square$

Corollary 4.2 (Natural moments and the first residual). *For $d \in \mathbb{N}_0$,*

$$\sum_{k=0}^n \lambda_{n,k}^{(q)} q^{kd} = \begin{cases} 1, & d = 0, \\ 0, & 1 \leq d \leq n, \\ (-1)^n q^{\binom{n+1}{2}} \begin{bmatrix} d-1 \\ n \end{bmatrix}_q, & d \geq n+1. \end{cases} \quad (28)$$

Proof. The first two cases follow from the zeros in (24). For $d \geq n + 1$,

$$(q^{d-n}; q)_n = \frac{(q; q)_{d-1}}{(q; q)_{d-n-1}},$$

and substitution into (26) gives the Gaussian coefficient. $\square$

Remark 4.3 (Fractional endpoint laws). For fixed s with $\Re s > 0$ and $s \notin \mathbb{N}$,

$$M_{n,q}(s) \sim q^{ns} \frac{(q^{1-s}; q)_\infty}{(q; q)_\infty} \quad (n \rightarrow \infty). \quad (29)$$

Thus a fractional power x^s is not annihilated, but its extrapolation error has a completely explicit geometric rate. Integer powers are exceptional because the finite product terminates at an exact zero.

4.2 Logarithmic powers by spectral differentiation

Differentiate (27) with respect to s . This gives closed responses for singular terms common in endpoint asymptotics.

Proposition 4.4 (Powers times logarithms). *For every integer $r \geq 0$,*

$$\sum_{k=0}^n \lambda_{n,k}^{(q)} (aq^k)^s (\log a + k \log q)^r = \frac{\partial^r}{\partial s^r} (a^s M_{n,q}(s)). \quad (30)$$

In particular, if $1 \leq d \leq n$, then

$$\sum_{k=0}^n \lambda_{n,k}^{(q)} (aq^k)^d \log(aq^k) = a^d M'_{n,q}(d), \quad (31)$$

where

$$M'_{n,q}(d) = (-1)^{d-1} (\log q) q^{d(2n-d+1)/2} \frac{(q; q)_{d-1} (q; q)_{n-d}}{(q; q)_n}. \quad (32)$$

Proof. Termwise differentiation is legitimate because the sum is finite. At $s = d$, the factor $q^s - q^d$ is the unique vanishing factor in (24). Differentiating that factor and splitting the remaining product below and above d gives

$$q^d \log q \prod_{\substack{1 \leq j \leq n \\ j \neq d}} (q^d - q^j) = (-1)^{d-1} (\log q) q^{d(2n-d+1)/2} (q; q)_{d-1} (q; q)_{n-d}.$$

Division by $(q; q)_n$ proves (32). $\square$

Remark 4.5. A filter that is exact on polynomials need not suppress logarithmic corrections attached to those powers. Formula (32) quantifies the leakage exactly. Higher s -derivatives introduce finite sums of q -harmonic terms $(1 - q^j)^{-r}$ and can be organized with complete Bell polynomials.

4.3 Mellin-transform interpretation

For a function on $(0, \infty)$ with Mellin transform

$$(\mathcal{M}f)(z) = \int_0^\infty f(x) x^{z-1} dx,$$

one has

$$\mathcal{M}[f(q^k x)](z) = q^{-kz} (\mathcal{M}f)(z).$$

Hence, on a common strip of convergence,

$$\mathcal{M}[\mathcal{G}_{n,q}f](z) = M_{n,q}(-z)(\mathcal{M}f)(z). \quad (33)$$

The geometric Lagrange filter is therefore a finite Mellin multiplier. Its spectral zeros are the same integer moments that the interpolation formula annihilates in physical space. This is the multiplicative analogue of an ordinary finite-difference stencil whose Fourier symbol has a prescribed zero at the origin.

4.4 The universal complete-symmetric residual

Let $v_0, \dots, v_n$ be arbitrary distinct nodes and let $h_r(z_0, \dots, z_m)$ denote the complete homogeneous symmetric polynomial of degree r . The following identity specializes particularly cleanly on a geometric comb.

Theorem 4.6 (Exact monomial residual). *Let ℓ_j be the Lagrange cardinal functions for $v_0, \dots, v_n$, and put $\Omega(x) = \prod_{j=0}^n (x - v_j)$. For every $r \geq 0$,*

$$x^{n+1+r} - \sum_{j=0}^n \ell_j(x) v_j^{n+1+r} = \Omega(x) h_r(x, v_0, \dots, v_n). \quad (34)$$

On $v_j = aq^j$,

$$h_r(x, a, aq, \dots, aq^n) = \sum_{s=0}^r x^s a^{r-s} \begin{bmatrix} n+r-s \\ n \end{bmatrix}_q. \quad (35)$$

Proof. The left side of (34) vanishes at every node, hence is divisible by Ω . Both sides are homogeneous of total degree $n+1+r$ in $x, v_0, \dots, v_n$. The quotient is identified from the generating function

$$\sum_{r \geq 0} h_r(x, v_0, \dots, v_n) t^r = \frac{1}{(1-xt) \prod_{j=0}^n (1-v_j t)}.$$

Equivalently, polynomial division of x^{n+1+r} by Ω gives the same recurrence. At geometric nodes,

$$\prod_{j=0}^n \frac{1}{1-aq^j t} = \sum_{m \geq 0} \begin{bmatrix} n+m \\ n \end{bmatrix}_q a^m t^m$$

by the reciprocal finite q -binomial theorem. Multiplication by $(1-xt)^{-1}$ and extraction of t^r proves (35). $\square$

Corollary 4.7 (Exact analytic interpolation tail). *Let $f(x) = \sum_{m \geq 0} c_m x^m$ converge absolutely in a disk containing the nodes and the target. Then*

$$f(x) - (\mathcal{I}_n f)(x) = \Omega_{n+1}(x) \sum_{r \geq 0} c_{n+1+r} \sum_{s=0}^r x^s a^{r-s} \begin{bmatrix} n+r-s \\ n \end{bmatrix}_q. \quad (36)$$

At $x = 0$ this becomes

$$(\mathcal{G}_{n,q} f)(a) - f(0) = (-1)^n a^{n+1} q^{\binom{n+1}{2}} \sum_{r \geq 0} c_{n+1+r} a^r \begin{bmatrix} n+r \\ n \end{bmatrix}_q. \quad (37)$$

Proof. Sum (34) over the monomial coefficients. Absolute convergence justifies rearrangement. Since $\Omega_{n+1}(0) = (-a)^{n+1} q^{\binom{n+1}{2}}$, moving the residual to the other side gives (37). $\square$

Corollary 4.8 (A disk bound). *Suppose f is analytic on $|z| < R$ and $|f(z)| \leq B$ on $|z| = R$. If $|a| < R$, then*

$$|(\mathcal{G}_{n,q} f)(a) - f(0)| \leq \frac{B q^{\binom{n+1}{2}} (|a|/R)^{n+1}}{(q; q)_n (1 - |a|/R)}. \quad (38)$$

Proof. Cauchy's estimate gives $|c_m| \leq BR^{-m}$, and $\begin{bmatrix} n+r \\ n \end{bmatrix}_q \leq 1/(q; q)_n$ for $0 < q < 1$. Apply these estimates to (37) and sum the geometric series. $\square$

5 Newton interpolation is Jackson calculus

5.1 Divided differences and iterated q -derivatives

Use the Jackson derivative convention

$$(D_q f)(x) = \frac{f(x) - f(qx)}{(1-q)x}. \quad (39)$$

It satisfies $D_q x^m = [m]_q x^{m-1}$.

Theorem 5.1 (Geometric divided differences are Jackson derivatives). *For every polynomial f , and more generally whenever all quantities are defined,*

$$f[a, aq, \dots, aq^k] = \frac{D_q^k f(a)}{[k]_q!}. \quad (40)$$

Equivalently,

$$\frac{D_q^k f(a)}{[k]_q!} = \frac{1}{a^k} \sum_{j=0}^k \frac{(-1)^j q^{-jk + \binom{j+1}{2}}}{(q; q)_j (q; q)_{k-j}} f(aq^j) \quad (41)$$

$$= \frac{1}{a^k (q; q)_k} \sum_{j=0}^k (-1)^j q^{-jk + \binom{j+1}{2}} \begin{bmatrix} k \\ j \end{bmatrix}_q f(aq^j). \quad (42)$$

Proof. Let $E_q f(x) = f(qx)$. Because $E_q x^{-1} = q^{-1} x^{-1} E_q$, iteration of (39) gives the operator identity

$$D_q^k = \frac{x^{-k}}{(1-q)^k} \prod_{r=0}^{k-1} (1 - q^{-r} E_q). \quad (43)$$

Expand the product with the finite q^{-1} -binomial theorem and use $\begin{bmatrix} k \\ j \end{bmatrix}_{q^{-1}} = q^{-j(k-j)} \begin{bmatrix} k \\ j \end{bmatrix}_q$. Division by $[k]_q! = (q; q)_k / (1-q)^k$ yields (42). On the other hand, the explicit formula for a divided difference is

$$f[a, aq, \dots, aq^k] = \sum_{j=0}^k \frac{f(aq^j)}{\prod_{r \neq j} (aq^j - aq^r)}.$$

The denominator calculation from Theorem 3.1, with $n = k$, is exactly (41). □

Corollary 5.2 (Finite Jackson–Newton formula). *The degree- n interpolant is*

$$(\mathcal{I}_n f)(x) = \sum_{k=0}^n \frac{D_q^k f(a)}{[k]_q!} \prod_{r=0}^{k-1} (x - aq^r). \quad (44)$$

For $f(x) = x^m$, the coefficient is

$$\frac{D_q^k x^m|_{x=a}}{[k]_q!} = \begin{bmatrix} m \\ k \end{bmatrix}_q a^{m-k}, \quad (45)$$

which recovers (10).

Remark 5.3. The phrase “ q -Taylor formula” can obscure the geometry. Formula (44) is literally the ordinary Newton formula on the nodes $a, aq, \dots$; Jackson calculus appears because those divided differences happen to close under dilation. This also fixes normalization ambiguities between the several q -derivative conventions in the literature.

5.2 Infinite Jackson–Newton series for analytic functions

Theorem 5.4 (Local analytic convergence). *Let $0 < q < 1$, and let $f(z) = \sum_{m \geq 0} c_m z^m$ be analytic in $|z| < R$. If $|a| < R$, define*

$$A_k = \frac{D_q^k f(a)}{[k]_q!} = \sum_{r \geq 0} c_{k+r} \begin{bmatrix} k+r \\ k \end{bmatrix}_q a^r. \quad (46)$$

Then

$$f(x) = \sum_{k=0}^{\infty} A_k \Phi_k(x) \quad (47)$$

locally uniformly for $|x| < R$. At the accumulation point,

$$f(0) = \sum_{k=0}^{\infty} (-a)^k q^{\binom{k}{2}} \frac{D_q^k f(a)}{[k]_q!}. \quad (48)$$

Proof. Fix $s < R$ and choose ρ with $\max(s, |a|) < \rho < R$. Cauchy's estimate gives $|c_m| \leq C\rho^{-m}$. Since $\begin{bmatrix} k+r \\ k \end{bmatrix}_q \leq 1/(q; q)_{\infty}$,

$$|A_k| \leq C_1 \rho^{-k}.$$

For $0 < s < R$ and $|x| \leq s$,

$$|\Phi_k(x)| \leq \prod_{r=0}^{k-1} (s + |a|q^r) \leq s^k \prod_{r=0}^{\infty} (1 + |a|q^r/s) = C_2(s)s^k.$$

Thus the series converges uniformly on $|x| \leq s$. The formal basis identity in [Corollary 2.6](#) then identifies its sum with f . At $x = 0$, $\Phi_k(0) = (-a)^k q^{\binom{k}{2}}$, which proves (48). $\square$

Remark 5.5 (No analyticity hypothesis can be silently dropped). The infinite series in (47) is much more rigid than the finite interpolation formula. The Fabius function is C^{∞} but flat at the origin and nowhere analytic; in [Section 6](#) we prove that its corresponding infinite Newton series diverges at every off-comb nonzero point.

5.3 The limits $q \rightarrow 1$ and $q \rightarrow 0$

Proposition 5.6 (Taylor limit). *For fixed k , as $q \rightarrow 1$,*

$$D_q^k f(a) \longrightarrow f^{(k)}(a), \quad [k]_q! \longrightarrow k!, \quad \Phi_k(x) \longrightarrow (x - a)^k.$$

Thus (44) tends termwise to the ordinary Taylor polynomial about a .

Proof. The first statement follows by repeated use of the standard Jackson derivative limit on a sufficiently differentiable function; the other two are immediate from their finite products. $\square$

At the opposite endpoint, $D_0 f(x) = (f(x) - f(0))/x$ is a tail-quotient operator. For $f(x) = \sum c_m x^m$,

$$\lim_{q \rightarrow 0} \frac{D_q^k f(a)}{[k]_q!} = \sum_{r \geq 0} c_{k+r} a^r, \quad \lim_{q \rightarrow 0} \Phi_k(x) = \begin{cases} 1, & k = 0, \\ (x - a)x^{k-1}, & k \geq 1. \end{cases}$$

The limiting expansion telescopes. It is not Hermite interpolation at a multiple node: the geometric rates retain ordered information about how the nodes approach zero. A genuine confluent theory requires a further renormalization and is listed as [Problem 11.3](#).

5.4 Neville, Richardson, and Romberg recurrences

Theorem 5.7 (Geometric Neville recurrence). *For $i, m \geq 0$, let $P_{i,m}$ be the interpolant characterized by*

$$P_{i,m}(aq^{i+r}) = f(aq^{i+r}) \quad (0 \leq r \leq m).$$

Then

$$P_{i,m}(x) = \frac{(x - aq^{i+m})P_{i,m-1}(x) - (x - aq^i)P_{i+1,m-1}(x)}{aq^i - aq^{i+m}}. \quad (49)$$

At $x = 0$, with $R_i^{(0)} = f(aq^i)$,

$$R_i^{(m)} = \frac{R_{i+1}^{(m-1)} - q^m R_i^{(m-1)}}{1 - q^m}, \quad R_0^{(n)} = (\mathcal{G}_{n,q}f)(a). \quad (50)$$

Proof. The numerator in (49) has degree at most m and takes the required values at both endpoint nodes and all shared interior nodes; uniqueness proves the formula. Substitute $x = 0$ and cancel aq^i . $\square$

Corollary 5.8 (Generalized Richardson cancellation). *Suppose approximations at scales $h_i = h_0 r^i$ have an expansion*

$$T(h) = L + c_1 h^\rho + c_2 h^{2\rho} + \dots$$

Apply (50) with $q = r^\rho$ to the data $T(h_i)$. After n levels, the first n error powers are annihilated exactly. Romberg quadrature is the special case in which the trapezoidal error uses even powers and hence $q = r^2$.

Remark 5.9. This recurrence is often numerically preferable to forming the closed weights. It also explains the historical term “deferred approach to the limit”: each new level eliminates one Mellin eigenmode of the asymptotic error.

6 Convergence at an accumulation point and divergence away from it

6.1 The zero row is a regular summability method

Theorem 6.1 (Regularity at the accumulation point). *Let $0 < q < 1$ and let $y_k \rightarrow L$. Then*

$$\sum_{k=0}^n \lambda_{n,k}^{(q)} y_k \longrightarrow L. \quad (51)$$

Equivalently, if f is continuous at 0, the interpolating polynomial through $(aq^k, f(aq^k))$, $0 \leq k \leq n$, satisfies

$$(\mathcal{I}_n f)(0) \longrightarrow f(0). \quad (52)$$

Proof. The rows reproduce constants by (28); their ℓ^1 norms are bounded by $K(q)$ by Theorem 3.7; and for each fixed k ,

$$|\lambda_{n,k}^{(q)}| = \frac{q^{\binom{n-k+1}{2}}}{(q; q)_k (q; q)_{n-k}} \longrightarrow 0.$$

These are precisely the Toeplitz regularity conditions. For completeness, subtract L , split the sum at a fixed K , make the tail small using the bounded row norm, and then let $n \rightarrow \infty$ in the finite head. $\square$

6.2 The limiting profile viewed from the finest tooth

Reindex the row by $r = n - k$. For each fixed r ,

$$\lambda_{n,n-r}^{(q)} \longrightarrow \mu_r^{(q)} := \frac{(-1)^r q^{\binom{r+1}{2}}}{(q; q)_\infty (q; q)_r}. \quad (53)$$

Euler's infinite theorem gives the generating function

$$\sum_{r=0}^{\infty} \mu_r^{(q)} z^r = \frac{(qz; q)_\infty}{(q; q)_\infty}, \quad \sum_{r=0}^{\infty} \mu_r^{(q)} = 1. \quad (54)$$

Thus the mass does not remain attached to any fixed physical node; it travels with the finest scale aq^n and approaches a signed, absolutely summable profile in the reversed index. This is the mechanism behind [Theorem 6.1](#).

6.3 Power, Puiseux, and logarithmic endpoint expansions

Suppose a germ has a convergent or asymptotic expansion

$$f(x) = L + \sum_r c_r x^{\alpha_r} (\log x)^{m_r}, \quad \Re \alpha_r > 0.$$

Theorems [4.1](#) and [4.4](#) give the filter of each term exactly. In particular, a nonintegral leading exponent α produces the rate $q^{n\alpha}$ times an explicit q -product, while an integral polynomial jet is deleted once n reaches its degree. The filter is therefore an endpoint asymptotic microscope: its decay rate distinguishes fractional regularity from an analytic integer jet.

Proposition 6.2 (A simple Hölder estimate). *If $|f(x) - f(0)| \leq C|x|^\alpha$ on the comb, with $\alpha > 0$, then*

$$|(\mathcal{G}_{n,q}f)(a) - f(0)| \leq C|a|^\alpha \sum_{k=0}^n |\lambda_{n,k}^{(q)}| q^{k\alpha}. \quad (55)$$

For fixed q and α , the right side is $O(q^{n\alpha})$.

Proof. The first statement is the triangle inequality. Reindex $r = n - k$. Since $(q; q)_m \geq (q; q)_\infty$ for every m , one has

$$\begin{aligned} q^{-n\alpha} \sum_{k=0}^n |\lambda_{n,k}^{(q)}| q^{k\alpha} &= \sum_{r=0}^n \frac{q^{\binom{r+1}{2} - r\alpha}}{(q; q)_{n-r} (q; q)_r} \\ &\leq \frac{1}{(q; q)_\infty^2} \sum_{r=0}^{\infty} q^{\binom{r+1}{2} - r\alpha} < \infty. \end{aligned}$$

The last series converges for every fixed α because its exponent is quadratic in r . This uniform bound proves the stated $O(q^{n\alpha})$ estimate. Dominated convergence also gives the sharper limit

$$q^{-n\alpha} \sum_{k=0}^n |\lambda_{n,k}^{(q)}| q^{k\alpha} \longrightarrow \sum_{r=0}^{\infty} |\mu_r^{(q)}| q^{-r\alpha}.$$

□

6.4 Superflat data force divergence off the comb

The favorable result at 0 has a nearly opposite counterpart everywhere else. It is useful to formulate it directly for data, without assuming that they initially came from a function.

Definition 6.3. A sequence $y_j \rightarrow L$ is q -superflat at its limit if

$$y_j - L = o(q^{mj}) \quad (j \rightarrow \infty) \quad (56)$$

for every $m \in \mathbb{N}$.

Theorem 6.4 (Exact convergence set for nontrivial superflat data). *Let $0 < q < 1$, $a \neq 0$, and let $y_j \rightarrow L$ be q -superflat. Assume $y_j \neq L$ for infinitely many j . Let*

$$P_n(x) = \sum_{k=0}^n A_k \Phi_k(x) \quad (57)$$

be the polynomial interpolating $P_n(aq^j) = y_j$ for $0 \leq j \leq n$. Then

$$\limsup_{k \rightarrow \infty} |A_k|^{1/k} = \infty. \quad (58)$$

Moreover:

1. $P_n(0) \rightarrow L$;
2. for each $m \geq 0$, $P_n(aq^m) = y_m$ once $n \geq m$;
3. for every $x \in \mathbb{C} \setminus (\{0\} \cup \{aq^m : m \geq 0\})$, the increments $A_n \Phi_n(x)$ fail to tend to zero, and the sequence $P_n(x)$ diverges; indeed it is unbounded along a subsequence.

Thus the convergence set is exactly the geometric comb together with its accumulation point.

Proof. Subtract L ; this changes only A_0 and reduces to $L = 0$. Suppose, contrary to (58), that $|A_k| \leq CR^{-k}$ for some $R > 0$. For every $0 < r < R$, the estimate

$$\sup_{|z| \leq r} |\Phi_k(z)| \leq r^k \prod_{j=0}^{\infty} (1 + |a|q^j/r)$$

shows that $g(z) = \sum_{k \geq 0} A_k \Phi_k(z)$ is analytic on $|z| < R$. For every sufficiently large j , $|aq^j| < R$, and all terms with $k > j$ vanish at aq^j . Hence

$$g(aq^j) = P_j(aq^j) = y_j.$$

If the first nonzero Taylor coefficient of g at 0 had order m , then $g(aq^j)/(aq^j)^m$ would tend to a nonzero constant. This contradicts q -superflatness. Therefore every Taylor coefficient vanishes, so g is identically zero near 0, contradicting the assumption that infinitely many y_j are nonzero. This proves (58).

Assertion 1 is [Theorem 6.1](#), and assertion 2 is interpolation. For a point x in assertion 3,

$$\Phi_n(x) = x^n \prod_{r=0}^{n-1} \left(1 - \frac{aq^r}{x}\right). \quad (59)$$

The infinite product tends to $(a/x; q)_\infty \neq 0$, because x is not a node. Hence $|\Phi_n(x)|^{1/n} \rightarrow |x| > 0$. Together with (58), this implies $\limsup |A_n \Phi_n(x)|^{1/n} = \infty$; the increments do not tend to zero. Finally, $|P_n - P_{n-1}| = |A_n \Phi_n|$ is unbounded along a subsequence, so at least one of each adjacent pair P_n, P_{n-1} is unbounded along a subsequence. $\square$

Remark 6.5 (A precise geometric Runge phenomenon). This is not the classical Runge phenomenon caused by equispaced nodes on a fixed interval. The comb is not dense away from 0. Superflat nonzero data cannot be the trace of a nonzero analytic germ at the accumulation point; the interpolation coefficients must therefore leave every exponential class. The instability is an analytic-continuation obstruction encoded in a Vandermonde problem.

Remark 6.6 (Derivative-enhanced interpolation). Adding finitely many Hermite conditions at any fixed collection of nonzero teeth does not remove the obstruction: an exponentially bounded coefficient sequence would still define an analytic germ with a superflat nonzero trace. A successful regularization must either change the approximation space, use a number of derivative conditions growing with n , or explicitly build flatness into the basis. Rvachev atoms are natural candidates for the last option.

7 The geometric-uniform family

7.1 Probability law, support, and refinement

Let $U_0, U_1, \dots$ be independent uniform random variables on $[0, 1]$ and set

$$X_q = (1 - q) \sum_{j=0}^{\infty} q^j U_j, \quad 0 < q < 1. \quad (60)$$

This is a geometric member of the infinite-convolution class studied by Jessen and Wintner [7]. The coefficients are positive and sum to 1, so $0 \leq X_q \leq 1$. Reflection of every U_j shows

$$X_q \stackrel{d}{=} 1 - X_q. \quad (61)$$

Let $F_q(x) = \mathbb{P}(X_q \leq x)$, extended by 0 to $x \leq 0$ and by 1 to $x \geq 1$. Removing the first coordinate gives the affine fixed-point law

$$X_q \stackrel{d}{=} (1 - q)U + qX'_q, \quad (62)$$

where X'_q is an independent copy. Consequently

$$F_q(x) = \int_0^1 F_q\left(\frac{x - (1 - q)u}{q}\right) du \quad (63)$$

$$= \frac{q}{1 - q} \int_{(x - (1 - q))/q}^{x/q} F_q(t) dt. \quad (64)$$

Differentiation gives

$$F'_q(x) = \frac{F_q(x/q) - F_q((x - (1 - q))/q)}{1 - q}. \quad (65)$$

Proposition 7.1 (Smooth compactly supported density). *The centered random variable*

$$Z_q = 2X_q - 1 = (1 - q) \sum_{j=0}^{\infty} q^j V_j, \quad V_j \sim \text{Unif}[-1, 1], \quad (66)$$

has an even C^∞ density u_q supported on $[-1, 1]$. With the Fourier convention $\widehat{f}(\xi) = \int_{\mathbb{R}} f(x) e^{-2\pi i \xi x} dx$,

$$\widehat{u}_q(\xi) = \prod_{j=0}^{\infty} \text{sinc}_\pi(2(1 - q)q^j \xi), \quad \text{sinc}_\pi(t) = \frac{\sin(\pi t)}{\pi t}. \quad (67)$$

Moreover

$$u_q(z) = \frac{1}{2} F'_q\left(\frac{z + 1}{2}\right). \quad (68)$$

Proof. The characteristic function factors by independence and gives (67). To see rapid decay, use $|\text{sinc}_\pi(t)| \leq \min(1, 1/(\pi|t|))$ on the first $N \asymp \log|\xi|/\log q$ factors. The logarithm of the resulting product is $-c_q(\log|\xi|)^2 + O_q(\log|\xi|)$, hence the product decreases faster than any inverse power. Fourier inversion then gives a C^∞ density. The support and evenness follow from (66); the density change of variables gives (68). $\square$

Remark 7.2. The repository's formal development treats a still broader geometric-uniform law, including negative ratios and uniqueness of the affine distributional fixed point. The positive-ratio case above is the one naturally matched to a one-sided interpolation comb.

7.2 Moments and Bernoulli cumulants

Write

$$\mu_n(q) = \mathbb{E}X_q^n. \quad (69)$$

Theorem 7.3 (Moment recurrence). *The moments satisfy $\mu_0(q) = 1$ and, for $n \geq 1$,*

$$(1 - q^n)\mu_n(q) = \sum_{k=0}^{n-1} \binom{n}{k} q^k (1 - q)^{n-k} \frac{\mu_k(q)}{n - k + 1}. \quad (70)$$

At $q = 1/2$ this reduces to

$$(2^n - 1)\mu_n(1/2) = \sum_{k=0}^{n-1} \binom{n}{k} \frac{\mu_k(1/2)}{n - k + 1}. \quad (71)$$

Proof. Raise (62) to the n th power, expand, and use $\mathbb{E}U^r = 1/(r + 1)$. Move the $k = n$ term to the left. At $q = 1/2$ every factor $q^k(1 - q)^{n-k}$ equals 2^{-n} , and multiplication by 2^n gives (71). $\square$

The centered moment generating function is

$$M_q(z) = \mathbb{E}e^{zZ_q} = \prod_{j=0}^{\infty} \frac{\sinh((1 - q)q^j z)}{(1 - q)q^j z}. \quad (72)$$

Using

$$\log \frac{\sinh z}{z} = \sum_{m \geq 1} \frac{2^{2m-1} B_{2m}}{m(2m)!} z^{2m},$$

we obtain the even cumulants

$$\kappa_{2m}(Z_q) = \frac{2^{2m} B_{2m}}{2m} \frac{(1 - q)^{2m}}{1 - q^{2m}}, \quad \kappa_{2m+1}(Z_q) = 0. \quad (73)$$

Thus the same geometric series that produces the interpolation denominators also produces the cumulant denominators. At $q = 1/2$,

$$\kappa_{2m}(Z_{1/2}) = \frac{B_{2m}}{2m(1 - 4^{-m})}. \quad (74)$$

Moments can be recovered from these cumulants by complete Bell polynomials, providing a second exact route parallel to (71).

7.3 The left boundary layer

For $0 \leq x \leq 1 - q$, the second term in (65) vanishes, so

$$F'_q(x) = \frac{1}{1-q} F_q(x/q). \quad (75)$$

Iteration gives the geometric derivative tower.

Lemma 7.4 (Boundary derivative tower). *If $m \geq 1$ and $0 \leq x \leq (1-q)q^{m-1}$, then*

$$F_q^{(m)}(x) = (1-q)^{-m} q^{-\binom{m}{2}} F_q(x/q^m). \quad (76)$$

Proof. Induct on m . The case $m = 1$ is (75). Differentiating the induction hypothesis contributes a factor q^{-1} , and the domain assumption ensures that the next use of (75) remains in the left boundary layer. $\square$

7.4 Comb samples are normalized moments

The ratio $q = 1/2$ is geometrically critical: the first uniform block has length $1 - q$, while the tail has length q , and these are equal exactly at $q = 1/2$. For every $q \leq 1/2$, however, the tooth q^n still lies inside the n th iterated boundary layer, which is enough for the following formula.

Theorem 7.5 (Geometric comb-moment identity). *For $0 < q \leq 1/2$ and every $n \geq 0$,*

$$F_q(q^n) = \frac{q^{\binom{n+1}{2}}}{(1-q)^n n!} \mu_n(q). \quad (77)$$

Proof. The case $n = 0$ is $F_q(1) = 1$. For $n \geq 1$, compact support and smoothness imply that F_q and all its derivatives vanish at 0. Since $q^n \leq (1-q)q^{n-1}$ when $q \leq 1/2$, Taylor's integral formula and Lemma 7.4 give

$$\begin{aligned} F_q(q^n) &= \frac{1}{(n-1)!} \int_0^{q^n} (q^n - t)^{n-1} F_q^{(n)}(t) dt \\ &= \frac{(1-q)^{-n} q^{-\binom{n}{2}}}{(n-1)!} \int_0^{q^n} (q^n - t)^{n-1} F_q(t/q^n) dt \\ &= \frac{q^{n^2 - \binom{n}{2}}}{(1-q)^n (n-1)!} \int_0^1 (1-u)^{n-1} F_q(u) du. \end{aligned}$$

By Fubini,

$$\int_0^1 (1-u)^{n-1} F_q(u) du = \frac{1}{n} \mathbb{E}(1 - X_q)^n.$$

Symmetry (61) changes the last moment to $\mu_n(q)$, and $n^2 - \binom{n}{2} = \binom{n+1}{2}$. $\square$

Remark 7.6 (What fails for $q > 1/2$). The formula is not expected to survive unchanged. The point q^n lies beyond the pure left boundary layer, so the reflected term in (65) enters before the n th differentiation. The resulting correction should be indexed by affine words recording boundary crossings. Deriving a closed word-sum or automaton formula is Problem 11.2.

7.5 The classical Fabius and Rvachev normalizations

Put

$$F = F_{1/2}, \quad X = X_{1/2}, \quad u = u_{1/2}.$$

Then F is the bounded Fabius function. On $0 \leq x \leq 1/2$,

$$F'(x) = 2F(2x), \quad F(1-x) = 1 - F(x).$$

The centered density u is Rvachev's up-function in the probability normalization. From (68),

$$u(z) = \begin{cases} F(z+1), & -1 \leq z \leq 0, \\ F(1-z), & 0 \leq z \leq 1, \\ 0, & |z| \geq 1. \end{cases} \quad (78)$$

Its Fourier transform is the canonical infinite sinc product

$$\widehat{u}(\xi) = \prod_{j=0}^{\infty} \text{sinc}_{\pi}(\xi/2^j). \quad (79)$$

The construction and these identities agree with the classical treatments of Fabius [4] and Arias de Reyna [1], and with the normalization used in the repository's primary exposition [11].

8 Exact Fabius transforms on the dyadic comb

8.1 Dyadic values and Mersenne products

Write

$$\mu_n = \mathbb{E}X^n, \quad \mathfrak{M}_n = \prod_{r=1}^n (2^r - 1), \quad \mathfrak{M}_0 = 1. \quad (80)$$

Theorems 7.3 and 7.5 become

$$(2^n - 1)\mu_n = \sum_{k=0}^{n-1} \binom{n}{k} \frac{\mu_k}{n - k + 1}, \quad (81)$$

$$F(2^{-n}) = 2^{-\binom{n}{2}} \frac{\mu_n}{n!}. \quad (82)$$

This moment formula is compatible with the exact dyadic arithmetic developed by Arias de Reyna [2] and with Haugland's exact evaluation algorithm [6]; here it is used as input to a second, interpolation-theoretic transform.

Since

$$(1/2; 1/2)_n = 2^{-\binom{n+1}{2}} \mathfrak{M}_n, \quad (83)$$

the zero-target row has the particularly simple form

$$\lambda_{n,k}^{(1/2)} = (-1)^{n-k} \frac{2^{\binom{k+1}{2}}}{\mathfrak{M}_k \mathfrak{M}_{n-k}}. \quad (84)$$

Thus every dyadic extrapolation coefficient is a rational expression in two complementary Mersenne products.

8.2 The endpoint extrapolant as a Mersenne-moment convolution

Let P_n interpolate F at $1, 1/2, \dots, 2^{-n}$ and put $E_n = P_n(0)$.

Theorem 8.1 (Exact endpoint convolution). *For every $n \geq 0$,*

$$E_n = \sum_{k=0}^n (-1)^{n-k} \frac{2^k \mu_k}{k! \mathfrak{M}_k \mathfrak{M}_{n-k}}. \quad (85)$$

The first values are

$$1, \quad 0, \quad -\frac{13}{27}, \quad \frac{1}{9}, \quad \frac{4418}{637875}, \quad -\frac{3218}{1318275}, \dots$$

Despite the initially nonmonotone behavior, $E_n \rightarrow 0$.

Proof. Substitute (82) and (84) into $E_n = \sum_k \lambda_{n,k}^{(1/2)} F(2^{-k})$. The powers combine as $2^{\binom{k+1}{2} - \binom{k}{2}} = 2^k$. Convergence follows independently from Theorem 6.1. $\square$

The same mechanism yields a quantitative bound throughout the geometric-uniform family.

Theorem 8.2 (Quadratic endpoint bound). *Let $0 < q \leq 1/2$, and let $P_{n,q}$ interpolate F_q at $1, q, \dots, q^n$. Then*

$$|P_{n,q}(0)| \leq \frac{\exp((1-q)^{-1})}{(q; q)_\infty^2} q^{\binom{n+1}{2} - \lfloor n^2/4 \rfloor}. \quad (86)$$

In particular the convergence is at least as fast as $q^{n^2/4 + O(n)}$, and hence faster than every fixed geometric rate.

Proof. Use the exact sample formula (77), the bound $0 \leq \mu_k(q) \leq 1$, and (16):

$$|P_{n,q}(0)| \leq \sum_{k=0}^n \frac{q^{\binom{n-k+1}{2} + \binom{k+1}{2}} (1-q)^{-k}}{(q; q)_k (q; q)_{n-k}} \frac{1}{k!}.$$

The exponent is

$$\binom{n+1}{2} - k(n-k) \geq \binom{n+1}{2} - \lfloor n^2/4 \rfloor.$$

Also $(q; q)_k, (q; q)_{n-k} \geq (q; q)_\infty$. Extract the smallest power of q and majorize the remaining sum by $\sum_{k \geq 0} (1-q)^{-k}/k! = \exp((1-q)^{-1})$. $\square$

8.3 Newton coefficients as a Gaussian moment transform

Write the Newton form

$$P_n(x) = \sum_{k=0}^n A_k \prod_{r=0}^{k-1} (x - 2^{-r}). \quad (87)$$

The coefficient A_k is independent of the final interpolation degree once the first $k+1$ nodes are present.

Theorem 8.3 (General geometric-uniform moment transform). *For $0 < q \leq 1/2$, let $A_n(q)$ be the Newton coefficient for the data $F_q(1), F_q(q), \dots$. Then*

$$A_n(q) = \frac{1}{(q; q)_n} \sum_{j=0}^n (-1)^j \begin{bmatrix} n \\ j \end{bmatrix}_{q^{-1}} \left(\frac{q}{1-q} \right)^j \frac{\mu_j(q)}{j!}. \quad (88)$$

The transform is triangular and invertible. If $B_k = (q; q)_k A_k(q)$ and $Q = q^{-1}$, then

$$\frac{\mu_n(q)}{n!} = \left(\frac{1-q}{q} \right)^n \sum_{k=0}^n (-1)^k Q^{\binom{n-k}{2}} \begin{bmatrix} n \\ k \end{bmatrix}_Q B_k. \quad (89)$$

Proof. Insert (77) into the explicit divided difference (42) with $a = 1$. The powers simplify by

$$q^{-jn + \binom{j+1}{2} + \binom{j+1}{2}} (1-q)^{-j} = q^{-j(n-j)} \left(\frac{q}{1-q} \right)^j,$$

and reciprocity $q^{-j(n-j)} \begin{bmatrix} n \\ j \end{bmatrix}_q = \begin{bmatrix} n \\ j \end{bmatrix}_{q^{-1}}$ proves (88). For inversion, factor the transform into a diagonal matrix and the Gaussian Pascal matrix, and use (8) with base Q ; direct multiplication gives (89). $\square$

Corollary 8.4 (Dyadic Gaussian and Mersenne forms). *For the Fabius function,*

$$A_n = \frac{1}{(1/2; 1/2)_n} \sum_{j=0}^n (-1)^j \begin{bmatrix} n \\ j \end{bmatrix}_2 \frac{\mu_j}{j!} \quad (90)$$

$$= 2^{\binom{n+1}{2}} \sum_{j=0}^n \frac{(-1)^j \mu_j}{j! \mathfrak{M}_j \mathfrak{M}_{n-j}}. \quad (91)$$

Moreover

$$E_n - E_{n-1} = (-1)^n 2^{-\binom{n}{2}} A_n \quad (n \geq 1). \quad (92)$$

Proof. Set $q = 1/2$ in (88). Then $q^{-1} = 2$ and $q/(1-q) = 1$. Use $\begin{bmatrix} n \\ j \end{bmatrix}_2 = \mathfrak{M}_n / (\mathfrak{M}_j \mathfrak{M}_{n-j})$ and (83) for the second form. Finally, the difference between two successive Newton partial sums at zero is $A_n \Phi_n(0) = (-1)^n 2^{-\binom{n}{2}} A_n$. $\square$

8.4 Generating-function factorization

Define

$$B(z) = \sum_{r \geq 0} \frac{z^r}{\mathfrak{M}_r} = (-z/2; 1/2)_\infty, \quad H(z) = \sum_{r \geq 0} \frac{(-1)^r \mu_r}{r! \mathfrak{M}_r} z^r. \quad (93)$$

The equality for B is Euler's infinite q -binomial theorem. The two exact convolutions above are equivalent to

$$\sum_{n \geq 0} 2^{-\binom{n+1}{2}} A_n z^n = B(z) H(z), \quad (94)$$

$$\sum_{n \geq 0} (-1)^n E_n z^n = B(z) H(2z). \quad (95)$$

These identities exhibit the Fabius Newton coefficients as a q -Borel-type transform of the moment generating function multiplied by a universal Euler product. They may be useful for saddle-point analysis of A_n .

8.5 Exact divergence of the Fabius interpolants

The Fabius function is positive on $(0, 1]$ and flat at 0: $F(x) = o(x^m)$ for every m . Therefore its dyadic samples satisfy the hypotheses of Theorem 6.4.

Corollary 8.5 (Exact convergence set for Fabius comb interpolation). *Let P_n interpolate F at $1, 1/2, \dots, 2^{-n}$. Then*

$$P_n(x) \text{ converges as } n \rightarrow \infty \iff x \in \{0, 1, 1/2, 1/4, \dots\}. \quad (96)$$

At 0 the limit is 0 and obeys the quadratic bound (86); at a sampled point 2^{-m} the sequence is eventually equal to $F(2^{-m})$; at every other nonzero complex point it is unbounded along a subsequence.

Proof. Apply Theorem 6.4 with $a = 1$, $q = 1/2$, and $y_j = F(2^{-j})$. $\square$

Corollary 8.6 (A rigorous upper scale for coefficient growth). *The Fabius Newton coefficients satisfy*

$$|A_n| \leq 2^{n^2/4+O(n)}. \quad (97)$$

On the other hand, $\limsup |A_n|^{1/n} = \infty$.

Proof. Combine (92) with the bounds for E_n and E_{n-1} in Theorem 8.2. The lower qualitative statement is part of Theorem 6.4. $\square$

Conjecture 8.7 (Quadratic coefficient exponent). *For the Fabius coefficients in (87),*

$$\lim_{n \rightarrow \infty} \frac{\log_2 |A_n|}{n^2} = \frac{1}{4}. \quad (98)$$

More generally, for $0 < q \leq 1/2$,

$$\lim_{n \rightarrow \infty} \frac{\log |A_n(q)|}{n^2 \log(1/q)} = \frac{1}{4}. \quad (99)$$

A weaker and probably more robust version replaces the limit by a limsup. The Gaussian coefficient in (88) is largest near $j = n/2$, where $q^{-j(n-j)} = q^{-n^2/4+O(1)}$; the open issue is to control alternating cancellation and the subquadratic moment factor. The order-80 experiment in Section 10 is consistent with (98).

9 The Rvachev Fourier product through the same filter

9.1 Tail products along a geometric comb

Put

$$s_q(\xi) = \text{sinc}_\pi(2(1-q)\xi), \quad S_{q,k}(\xi) = \prod_{j=0}^{k-1} s_q(q^j \xi), \quad S_{q,0} = 1. \quad (100)$$

The infinite product (67) satisfies

$$\widehat{u}_q(q^k \xi) = \frac{\widehat{u}_q(\xi)}{S_{q,k}(\xi)}. \quad (101)$$

Thus values of $\widehat{u}_q$ on a geometric frequency comb are exactly the tails of its defining product.

Theorem 9.1 (Reciprocal sinc-product filter). *Let $0 < q < 1$ and suppose $\widehat{u}_q(\xi) \neq 0$. Define*

$$R_{n,q}(\xi) = \sum_{k=0}^n \frac{\lambda_{n,k}^{(q)}}{S_{q,k}(\xi)}. \quad (102)$$

Then

$$R_{n,q}(\xi) \longrightarrow \frac{1}{\widehat{u}_q(\xi)}. \quad (103)$$

More precisely,

$$\left| R_{n,q}(\xi) - \frac{1}{\widehat{u}_q(\xi)} \right| \leq \frac{q^{\binom{n+1}{2}}}{|\widehat{u}_q(\xi)|(q; q)_n} e^{2\pi|\xi|} \frac{(2\pi|\xi|)^{n+1}}{(n+1)!}. \quad (104)$$

Proof. Apply the geometric filter to the entire function of t

$$h_\xi(t) = \widehat{u}_q(t\xi) = \mathbb{E} e^{-2\pi i t \xi Z_q}.$$

Since $h_\xi(0) = 1$, Theorem 6.1 already gives $\sum_k \lambda_{n,k} h_\xi(q^k) \rightarrow 1$. By (101), this sum is $\widehat{u}_q(\xi) R_{n,q}(\xi)$, proving the limit. For the quantitative bound, the Taylor coefficients of h_ξ satisfy $|c_m| \leq (2\pi|\xi|)^m/m!$ because $|Z_q| \leq 1$. Formula (37) with $a = 1$ and $\lceil n+r \rceil_q \leq 1/(q; q)_n$ gives

$$|\widehat{u}_q(\xi) R_{n,q}(\xi) - 1| \leq \frac{q^{\binom{n+1}{2}}}{(q; q)_n} \sum_{m=n+1}^{\infty} \frac{(2\pi|\xi|)^m}{m!}.$$

The exponential-tail bound $\sum_{m \geq n+1} t^m/m! \leq e^t t^{n+1}/(n+1)!$ completes the proof. $\square$

Remark 9.2 (Zeros and regularized reciprocals). The hypothesis excludes the union of zero lattices $2(1-q)q^j\xi \in \mathbb{Z} \setminus \{0\}$. Near a zero, the individual prefix reciprocals are badly conditioned. Multiplying by the known vanishing factor, or differentiating the filtered identity to reconstruct the finite part, should produce stable formulas for residues and zero multiplicities; see [Problem 11.5](#).

9.2 Moment interpretation of the residual

Expanding the characteristic function gives

$$\widehat{u}_q(t\xi) = \sum_{m=0}^{\infty} \frac{(-2\pi i\xi)^m}{m!} \mathbb{E} Z_q^m t^m. \quad (105)$$

Substitution into (37) expresses the error in (103) as a Gaussian transform of the centered moments. Odd moments vanish, so the actual leading term is the first even degree above n . The reciprocal product identity is therefore the Fourier counterpart of the Fabius moment transform: the same filter reads moments from the CDF comb and cancels moments in the characteristic-function comb.

9.3 Exact synthesis of geometric cardinals by shifted up-functions

The adjacent repository report on the Lagrange–Legendre–Rvachev loop proves the following finite synthesis theorem. Let $u = \text{up}$ be the classical up density, let

$$\mathbf{M}_u(z) = \int_{-1}^1 e^{zx} u(x) dx, \quad \mathcal{D}_u = \mathbf{M}_u(D)^{-1} \quad (106)$$

act on polynomials, and let p have degree at most d . With $M = 2^d$,

$$p(x) = \frac{1}{M} \sum_{|r| < 2M} (\mathcal{D}_u p)(r/M) u(x - r/M), \quad -1 \leq x \leq 1. \quad (107)$$

See [12] for the proof, nullspace, minimal dictionaries, and conditioning analysis.

Applying (107) to the geometric cardinal functions gives an exact three-stage transform

$$\{f(q^j)\}_{j=0}^n \xrightarrow{\text{Gaussian Lagrange}} \mathcal{I}_n f \xrightarrow{\mathbf{M}_u(D)^{-1}} \mathcal{D}_u \mathcal{I}_n f \xrightarrow{\text{dyadic sampling}} \text{up-atom coefficients}.$$

Explicitly, for $M = 2^n$,

$$(\mathcal{I}_n f)(x) = \sum_{|r| < 2M} C_{n,r}(f) u(x - r/M), \quad (108)$$

where

$$C_{n,r}(f) = \frac{1}{M} \sum_{j=0}^n f(aq^j) (\mathcal{D}_u \ell_{n,j})(r/M). \quad (109)$$

The dependence on the data is entirely finite and linear.

To make the q -structure in (109) explicit, let $Q_m(t) = \mathcal{D}_u(t^m)$ be the Appell-deconvolved monomials. Define the excluded elementary symmetric functions

$$e_s^{(j)} = e_s(1, q, \dots, \widehat{q^j}, \dots, q^n). \quad (110)$$

They have the Gaussian formula

$$e_s^{(j)} = \sum_{r=0}^s (-1)^r q^{jr + \binom{s-r}{2}} \begin{bmatrix} n+1 \\ s-r \end{bmatrix}_q. \quad (111)$$

Therefore

$$(\mathcal{D}_u \ell_{n,j})(t) = \beta_{n,j} \sum_{m=0}^n (-a)^{n-m} e_{n-m}^{(j)} Q_m(t). \quad (112)$$

This is an explicit Gaussian–Appell transmutation from geometric sample data to compactly supported C^∞ atoms.

Proof of (111) and (112). For the full set, $e_s(1, q, \dots, q^n) = q^{\binom{s}{2}} \left[\begin{smallmatrix} n+1 \\ s \end{smallmatrix} \right]_q$ by (1). Removing q^j gives the recurrence $e_s^{\text{full}} = e_s^{(j)} + q^j e_{s-1}^{(j)}$; iteration proves (111). Expanding $\prod_{r \neq j} (x - aq^r)$ in monomials and using $\ell_{n,j} = \beta_{n,j} \prod_{r \neq j} (x - aq^r)$ gives (112) after applying $\mathcal{D}_u$ termwise. $\square$

9.4 The additive Thue–Morse companion

Let

$$X^{(N)} = \sum_{j=1}^N 2^{-j} U_j. \quad (113)$$

The CDF of a sum of independent uniforms with widths w_j is a truncated-power spline. For the dyadic widths, all subset sums form the uniform grid $m/2^N$ and their inclusion–exclusion signs become the Thue–Morse signs. Thus

$$\mathbb{P}(X^{(N)} \leq x) = \frac{2^{N(N+1)/2}}{N!} \sum_{m=0}^{2^N-1} (-1)^{s_2(m)} \left(x - \frac{m}{2^N} \right)_+^N, \quad (114)$$

where $s_2(m)$ is the binary digit sum. Differentiating gives the density with power $N - 1$ and factor $(N - 1)!^{-1}$.

The Gaussian comb filter and the Thue–Morse spline are two different projections of the same hierarchy of scales:

- the Lagrange filter keeps the scale index j and uses only $n + 1$ multiplicative nodes q^j ; Gaussian coefficients encode its exact moment cancellations;
- the finite convolution expands all 2^N subsets of dyadic scales; subset sums become additive knots and parity produces Thue–Morse cancellation.

At $q = 1/2$, the products $(1 - 2^{-j})$, the Mersenne factors $(2^j - 1)$, Gaussian coefficients at bases $1/2$ and 2 , and the binary subset lattice all coexist. This is why the Fabius/Rvachev system is not merely an example of the general geometric theory: it is the point at which multiplicative interpolation, additive splines, and binary arithmetic lock together exactly.

10 Numerical experiments and exact checks

10.1 Reproducibility

The archive contains `geometric_comb_experiments.py`. It uses `exact_fractions.Fraction` arithmetic for all finite $q = 1/2$ identities and high-precision `mpmath` arithmetic only for infinite sinc products. Running

```
python geometric_comb_experiments.py --output-dir output --max-order 80
```

regenerates the CSV tables and all figures. The script independently checks:

- the closed barycentric weights against literal node products;
- reproduction of constants, annihilation of the first n moments, and the first Gaussian residual;

- the identity between Jackson derivatives and geometric divided differences for an exact rational polynomial;
- the full nonintegral Mellin symbol at $q = 2/3$ to 65 decimal places;
- the two independent Gaussian and Mersenne formulas for the first 21 Fabius Newton coefficients;
- the endpoint convolution against direct Lagrange evaluation;
- the Newton increment identity (92).

No floating-point observation is used in the proofs above. The figures serve to reveal scale, transients, and plausible asymptotic refinements.

10.2 Exact endpoint values

Table 1: The first Fabius geometric-comb extrapolants $E_n = P_n(0)$.

n	exact value	decimal value
0	1	1.000000000000
1	0	0
2	$-13/27$	$-4.814814814815 \times 10^{-1}$
3	$1/9$	$1.111111111111 \times 10^{-1}$
4	$4418/637875$	$6.926121889085 \times 10^{-3}$
5	$-3218/1318275$	$-2.441068820997 \times 10^{-3}$
6	$92322532/1648153544625$	$5.601573488167 \times 10^{-5}$
7	$20600852/3322468256625$	$6.200466162144 \times 10^{-6}$
8	$-35299061419547/204161105975753390625$	$-1.728980711132 \times 10^{-7}$
9	$-943246141903/409122843739646990625$	$-2.305532815721 \times 10^{-9}$
10	$227162469903601505641/3602957316310515949176818109375$	$6.304889288453 \times 10^{-11}$

The first four degrees do not yet display the asymptotic regime. Thereafter the quadratic exponent is unmistakable.

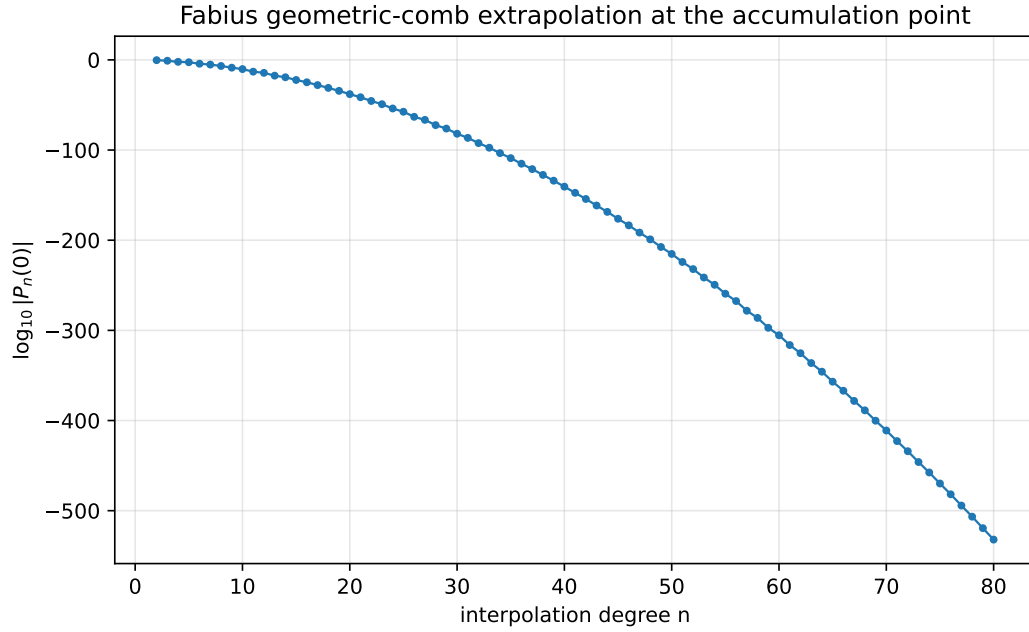


Figure 1: Exact rational extrapolation errors at the accumulation point. The roughly parabolic graph on a logarithmic scale is consistent with the proved $2^{-n^2/4+O(n)}$ bound.

10.3 Divergence away from the comb

At $x = 1/3$, the first few interpolants look innocuous:

$$P_5(1/3) \approx 0.1746, \quad P_{10}(1/3) \approx 0.2503.$$

The later values are enormous:

n	$\log_{10}(1 + P_n(1/3))$
15	4.1921
20	12.3248
30	39.4610
40	80.7599
60	206.0893
80	392.7146

This does not merely suggest divergence; [Corollary 8.5](#) proves it at every off-comb nonzero point.

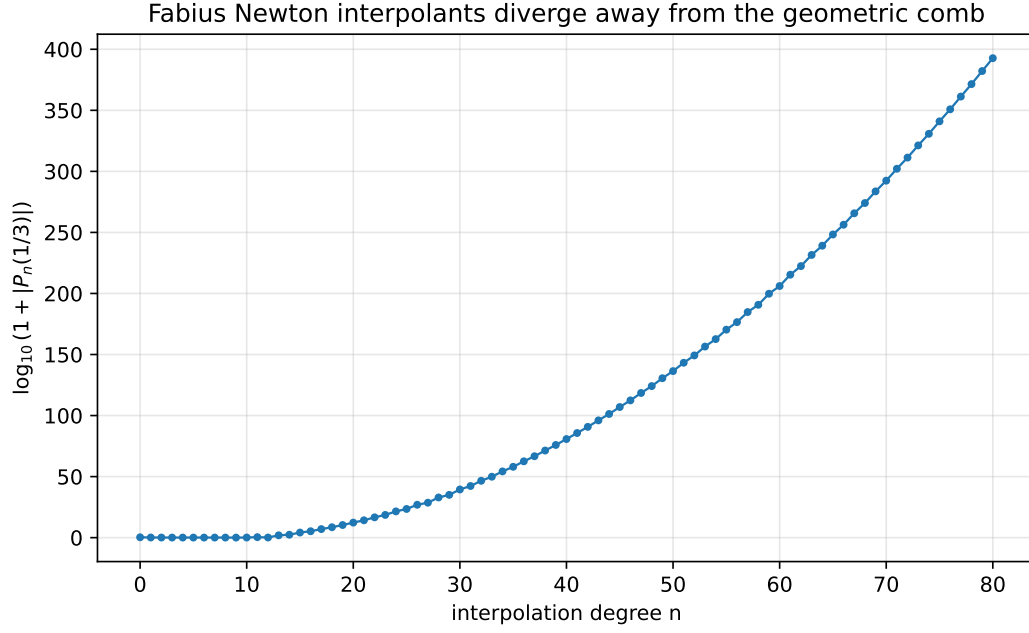


Figure 2: Growth of the exact rational interpolants at $x = 1/3$. The small values through degree 10 are a long transient, not convergence.

10.4 Quadratic growth of the Newton coefficients

The normalized quantity $n^{-2} \log_2 |A_n|$ rises slowly toward $1/4$ through order 80. The slow approach is compatible with logarithmic and periodic subquadratic corrections inherited from the endpoint asymptotics of the Fabius function.

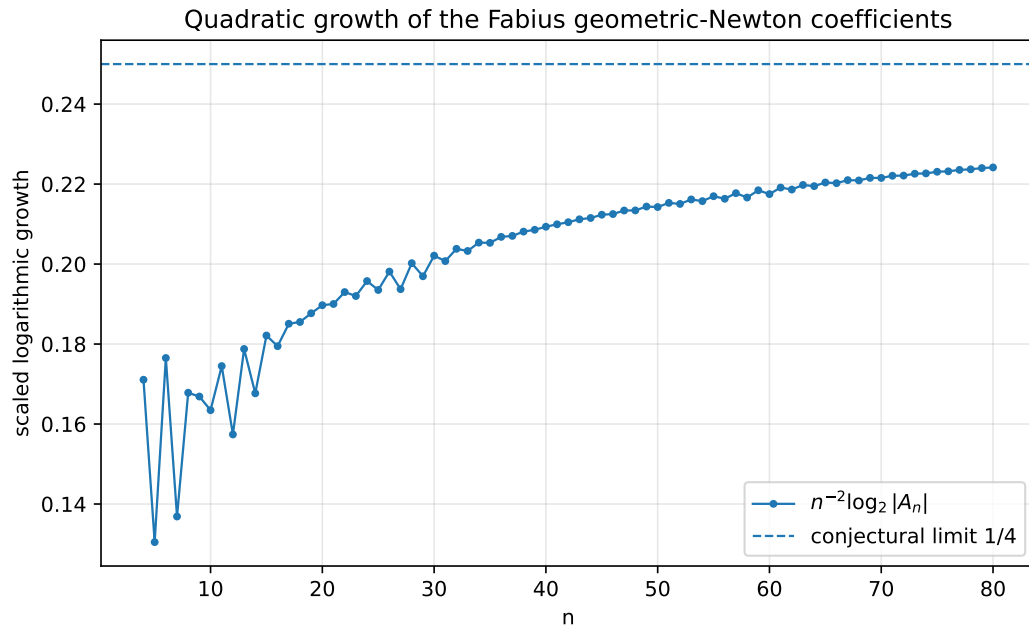


Figure 3: Numerical evidence for [Conjecture 8.7](#). The dashed line is $1/4$; it is not fitted to the data.

10.5 Bounded row variation

[Theorem 3.7](#) predicts saturation at a finite $K(q)$ for every fixed $q < 1$. The saturation is rapid for $q = 1/4$ and $q = 1/2$, while $q = 3/4$ requires more levels and reaches a much larger condition number.

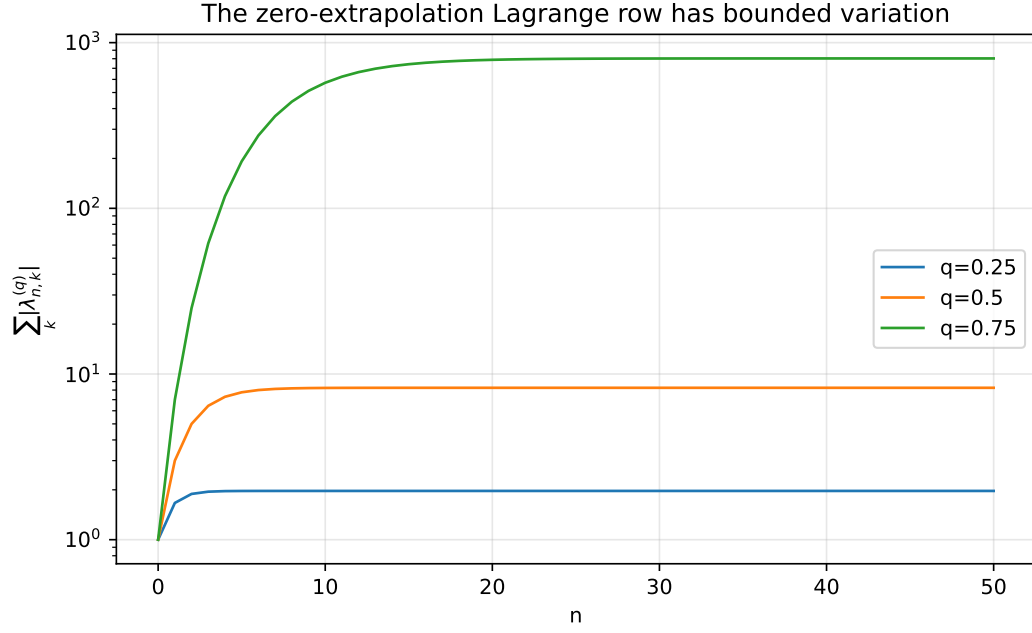


Figure 4: The exact row norm $(-q; q)_n / (q; q)_n$ on a logarithmic vertical scale. Fixed-ratio stability and the singular $q \rightarrow 1$ limit are both visible.

10.6 Reciprocal Rvachev product

For $q = 1/2$ and $\xi = 0.37$,

$$\begin{aligned}\hat{u}(0.37) &\approx 0.7319398291936895108374260130, \\ \hat{u}(0.37)^{-1} &\approx 1.3662325236510323730502298892.\end{aligned}$$

The degree-28 reciprocal filter has absolute error about 1.0×10^{-100} at 100-digit working precision.

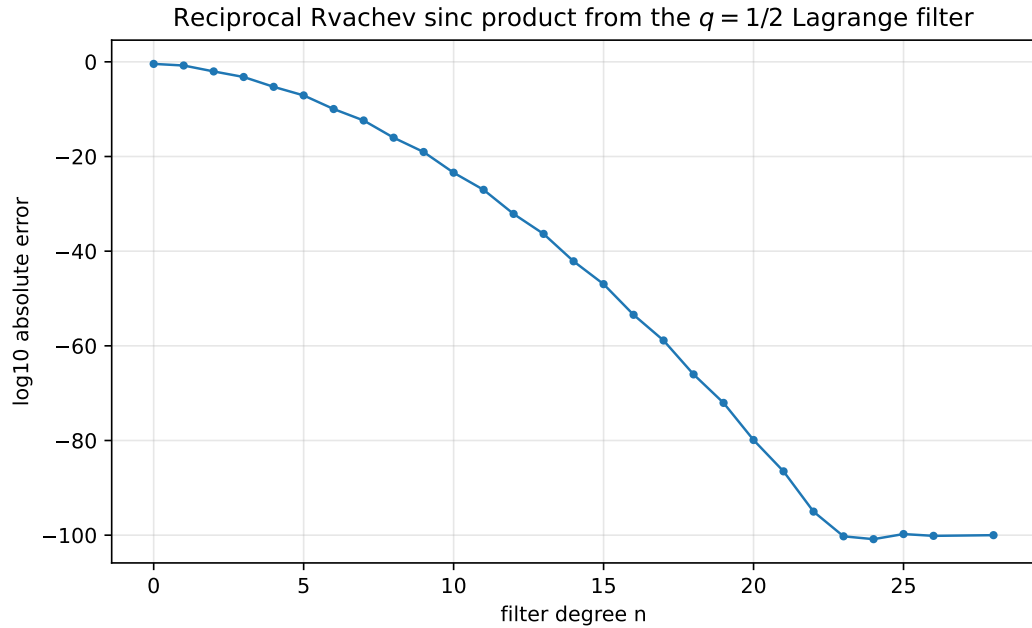


Figure 5: High-precision error in the reciprocal-product approximant (102). The combination of the Gaussian factor $q^{\binom{n+1}{2}}$ and the factorial Taylor tail gives extremely rapid convergence.

11 Research frontiers

The problems below are ordered from sharply formulated local questions to broader structural programs. None is needed for the proved results.

11.1 Asymptotics of the Fabius Newton coefficients

Conjecture 8.7 asks only for the leading quadratic exponent. A more informative target is a saddle-point expansion of (90) or (94). The Gaussian coefficient favors $j \approx n/2$, while $\mu_j/j!$ has its own lower-Lambert saddle and periodic correction. The interaction should produce a two-scale saddle problem.

Problem 11.1 (Full coefficient asymptotic). *Determine functions $c_2, c_1, c_{\log}$ and a bounded oscillatory term Ψ such that*

$$\log |A_n| = c_2 n^2 + c_1 n \log n + c_{\log} n + \Psi(\vartheta_n) + o(1), \quad (115)$$

with an explicit Lambert-type phase ϑ_n . Decide whether sign changes and near-cancellations are controlled by the same periodic function that occurs in the sharp Fabius endpoint asymptotic.

A first rigorous step would be matching upper and lower bounds $2^{n^2/4+o(n^2)}$. The upper bound is already in [Corollary 8.6](#); a lower bound must prevent exponential-scale cancellation in the alternating central block.

11.2 Boundary-crossing formulas for $q > 1/2$

Problem 11.2 (The supercritical comb-moment law). *For $1/2 < q < 1$, derive a closed formula for $F_q(q^n)$ in terms of moments and finite affine words generated by*

$$x \mapsto x/q, \quad x \mapsto (x - (1 - q))/q.$$

A satisfactory formula should specialize to (77) when no word crosses the reflected boundary, and should make the breakpoints in q explicit.

The number of admissible boundary words may change when polynomial relations among q -powers are crossed. This suggests a piecewise-algebraic structure related to beta-expansions, but with continuous uniform digits rather than Bernoulli digits.

11.3 Confluent limits and roots of unity

Problem 11.3 ($q \rightarrow 0$ confluent calculus). *Find a renormalized cardinal and Newton basis whose $q \rightarrow 0$ limit separates the infinitely many nodes collapsing at 0. Determine whether the correct limit is a triangular mixture of value, derivative, and tail-quotient functionals.*

Problem 11.4 (Cyclotomic geometric interpolation). *When q is a primitive m th root of unity, replace repeated nodes by a canonical Hermite data set and factor the resulting confluent Vandermonde matrix into cyclotomic analogues of P_q and $D_{a,q}$. The denominator-free operator $\prod_{j=1}^n (E_q - q^j)$ should survive as a periodic difference operator even when normalized Lagrange weights do not.*

These limits may connect the present theory to divided powers, quantum groups at roots of unity, and cyclic q -difference equations.

11.4 Regularization near Fourier zeros

Problem 11.5 (Filtered finite parts at sinc zeros). *Let ξ_0 be a zero of $\widehat{u}_q$ of known multiplicity m . Construct from $R_{n,q}$ and its first m derivatives a sequence converging to the finite part of $1/\widehat{u}_q$ at ξ_0 , with a bound uniform in a shrinking neighborhood of the zero.*

Because every zero comes from explicit finite sinc factors, one can first divide out the known local factor and then apply the geometric filter to the nonvanishing tail. The main issue is numerical cancellation when several zero lattices overlap, as they do extensively at $q = 1/2$.

11.5 Exact Lebesgue functions below the finest node

For $x = aq^\alpha$ with real $\alpha > n$, the cardinal signs are $(-1)^{n-j}$, so the Lebesgue function is the explicit positive sum below. The formula extends continuously to $\alpha = n$, where the target is the finest node and the Lebesgue function equals 1.

$$\Lambda_n(aq^\alpha) = (q^{\alpha-n}; q)_{n+1} \sum_{j=0}^n \frac{q^{\binom{n-j+1}{2}}}{(1 - q^{\alpha-j})(q; q)_j (q; q)_{n-j}}. \quad (116)$$

At $\alpha = \infty$ it reduces to (19). It is natural to ask whether the finite sum has a single basic-hypergeometric closed form and whether Λ_n is monotone on $[0, aq^n]$.

Conjecture 11.6 (Maximum below the comb). *For $0 < q < 1$ and $a > 0$,*

$$\max_{0 \leq x \leq aq^n} \Lambda_n(x) = \Lambda_n(0) = \frac{(-q; q)_n}{(q; q)_n}. \quad (117)$$

Low-degree symbolic and numerical checks support the claim, but a proof appears to require a variation-diminishing or total-positivity argument rather than termwise monotonicity.

11.6 Flatness-adapted approximation spaces

Polynomial interpolation fails off the comb for a structural reason. The basis must be changed, not merely evaluated more accurately. Classical geometric-knot spline constructions [9] provide one nearby approximation-theoretic model, while Rvachev atoms add exact C^∞ compact support and endpoint flatness.

Problem 11.7 (Rvachev-preconditioned comb interpolation). *Construct finite-dimensional spaces generated by scale-dependent Rvachev atoms, for example*

$$u(2^j x - r), \quad 0 \leq j \leq n,$$

that interpolate the Fabius dyadic data and incorporate endpoint flatness. Determine whether one can obtain convergence on $[0, 1]$ while retaining exact rational or algebraic coefficient formulas.

The deconvolution formula (112) supplies an exact preconditioner for each finite polynomial interpolant, but it cannot remove the divergence because it reproduces the same polynomial. A genuinely new scheme must let the atom space grow in smoothness or scale, rather than merely re-expressing $\mathbb{P}_n$.

Problem 11.8 (Growing Hermite data). *At the tooth 2^{-j} , include derivatives through an order r_j that grows with j . Find the minimal growth of r_j for which the Hermite interpolants can converge to F on a nontrivial interval. Relate the threshold to the sharp derivative and endpoint asymptotics of F .*

11.7 Multivariate geometric meshes

Schoenberg's one-dimensional problem has a triangular-mesh extension due to Lee and Phillips [10]. The present operator and probability viewpoints suggest several further versions.

Problem 11.9 (Tensor and simplex filters). *For nodes $(a_1 q_1^{j_1}, \dots, a_d q_d^{j_d})$, develop tensor and total-degree Lagrange filters with explicit multivariate Mellin symbols. On a simplex-like geometric mesh, identify the correct multivariate Gaussian coefficients and complete homogeneous residuals. Apply them to joint geometric-uniform laws and tensor products of up-functions.*

The tensor case is immediate algebraically but may suffer severe anisotropic conditioning. The simplex case should involve q -multinomial coefficients and may admit a probabilistic interpretation through flags of coordinate subspaces.

11.8 Adaptive ratios and asymptotic optimization

The exact error factors make ratio selection an optimization problem. Small q gives powerful Gaussian suppression but spreads the first nodes; q near 1 resolves a local germ but causes the condition number (22) to explode.

Problem 11.10 (Optimal q under noise). *Suppose the data $f(aq^k)$ have absolute or relative noise of known scale and f belongs to a specified analytic, Gevrey, or endpoint-asymptotic class. Minimize a rigorous sum of truncation and amplified-noise bounds over both q and n . Determine whether the optimum has a universal scaling law as the noise tends to zero.*

For the Fabius function, the superflat signal and the large off-comb coefficients make this a non-standard regularization problem; the optimum for evaluating $F(0)$ need not resemble the optimum for reconstructing $F(x)$ at $x > 0$.

11.9 Formal verification roadmap

The finite algebra is particularly well suited to Lean because most identities can be stated over a commutative ring before any division is introduced. A productive dependency order is:

1. prove the denominator-free nodal product and geometric Vandermonde factorization;
2. define the Gaussian Pascal matrix and prove (8);
3. derive the barycentric denominator and general-target cardinal after adding field hypotheses;
4. define the dilation polynomial and prove the full polynomial identity (17);
5. obtain natural moment annihilation by evaluation, then extend to the analytic Mellin symbol in a complex-power layer;
6. formalize Jackson divided differences first for polynomials, then for functions with iterated q -derivatives;
7. prove the row-norm identity and Toeplitz regularity using summable reversed rows;
8. reuse the repository's geometric-uniform law to prove (77);
9. instantiate at $q = 1/2$ and connect to existing Fabius moments and Mersenne products;
10. formalize the superflat divergence theorem using a local power-series contradiction;
11. feed the existing sinc-product theorem through the analytic filter to obtain Theorem 9.1.

The distinction between denominator-free polynomial statements and normalized field statements should be maintained throughout. It will make the later root-of-unity and confluent developments substantially easier.

12 Conclusions

Interpolation on $a, aq, aq^2, \dots$ is one of the places where ordinary numerical analysis and q -analysis are literally the same theory written in different coordinates. The nodal polynomial is a q -Pochhammer symbol, the Vandermonde factorization is Gaussian Pascal, the Newton coefficients are Jackson derivatives, and the extrapolation row is a finite polynomial in the dilation operator. Its Mellin symbol

$$q^{ns} \frac{(q^{1-s}; q)_n}{(q; q)_n}$$

contains the entire exactness theory, including fractional and logarithmic endpoint terms.

The accumulation point has a subtle status. Fixed-ratio extrapolation to zero is stable in the Toeplitz sense because the row norm tends to $(-q; q)_\infty / (q; q)_\infty$. Yet interpolation away from the comb is not a reconstruction method for arbitrary smooth functions. For nonzero superflat data the Newton coefficients grow beyond every exponential scale, and the interpolants diverge everywhere except on the sampled comb and at its limit. The Fabius function realizes this theorem in an exact arithmetic setting.

The probabilistic family $X_q = (1 - q) \sum q^j U_j$ completes the bridge. Its moments carry $1 - q^n$ denominators, its CDF samples on the comb become normalized moments for $q \leq 1/2$, and its centered density has the same geometric sinc product that defines Rvachev's up-function at $q = 1/2$. Accordingly, the Gaussian filter acts on both sides: it converts Fabius samples into Mersenne moment transforms, and it converts Rvachev tail products into a reciprocal Fourier expansion.

The most promising next steps are a full saddle asymptotic for the Fabius Newton coefficients, a boundary-word formula for $q > 1/2$, and approximation spaces that encode flatness from the outset. Those problems sit precisely at the interface of q -series, endpoint asymptotics, compactly supported smooth atoms, and formally verified analysis.

A Selected finite identities

For quick reference, the principal finite formulas are collected here.

$$\begin{aligned} \Omega_{n+1}(x) &= x^{n+1} (a/x; q)_{n+1}, \\ \Phi_k(x) &= (-a)^k q^{\binom{k}{2}} (x/a; q^{-1})_k, \\ \Phi_k(aq^j) &= (-a)^k q^{\binom{k}{2}} (q; q)_k \begin{bmatrix} j \\ k \end{bmatrix}_q, \\ \beta_{n,j} &= \frac{(-1)^j q^{-jn + \binom{j+1}{2}}}{a^n (q; q)_j (q; q)_{n-j}}, \\ \lambda_{n,j}^{(q)} &= \frac{(-1)^{n-j} q^{\binom{n-j+1}{2}}}{(q; q)_j (q; q)_{n-j}}, \\ \sum_{j=0}^n \lambda_{n,j}^{(q)} z^j &= \frac{\prod_{r=1}^n (z - q^r)}{(q; q)_n}, \\ f[a, aq, \dots, aq^k] &= \frac{D_q^k f(a)}{[k]_q!}, \\ \sum_{j=0}^n |\lambda_{n,j}^{(q)}| &= \frac{(-q; q)_n}{(q; q)_n}, \end{aligned}$$

$$M_{n,q}(s) = q^{ns} \frac{(q^{1-s}; q)_n}{(q; q)_n},$$

$$F_q(q^n) = \frac{q^{\binom{n+1}{2}}}{(1-q)^{n!}} \mathbb{E} X_q^n \quad (0 < q \leq 1/2),$$

$$F(2^{-n}) = 2^{-\binom{n}{2}} \frac{\mu_n}{n!},$$

$$A_n = \frac{1}{(1/2; 1/2)_n} \sum_{j=0}^n (-1)^j \begin{bmatrix} n \\ j \end{bmatrix}_2 \frac{\mu_j}{j!}.$$

B Pseudocode for exact dyadic computation

The following high-level algorithm uses rational arithmetic throughout.

```

mu[0] = 1
M[0] = 1
for n = 1..N:
    mu[n] = (sum_{k=0}^{n-1} binom(n,k)*mu[k]/(n-k+1)) / (2^n-1)
    M[n] = M[n-1]*(2^n-1)

for n = 0..N:
    Fdyadic[n] = 2^(-n*(n-1)/2) * mu[n] / n!
    A[n] = 2^(n*(n+1)/2) *
        sum_{j=0}^n (-1)^j * mu[j] / (j!*M[j]*M[n-j])
    E[n] = sum_{j=0}^n (-1)^(n-j) * 2^j * mu[j]
        / (j!*M[j]*M[n-j])

```

The attached Python implementation additionally verifies all transforms by independent formulas, evaluates Newton partial sums at rational targets, and computes the high-precision Fourier-product experiment.

C Source map

The repository files most directly related to this report are:

- Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/exponents-and-q-series/q_pochhammer_q_binomial_monograph/q_pochhammer_q_binomial_monograph.tex, which supplies the initial geometric Newton and exact-filter formulas;
- Analysis/FabiusFunction/docs/Fabius_Function_and_Rvachev_Up/Fabius_Function_and_Rvachev_Up.tex, the primary human-readable synthesis of the formal Fabius development and geometric-uniform law;
- Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/representations/lagrange_rvachev_loop_report_v3/lagrange_rvachev_loop.tex, which develops finite Lagrange/Legendre/up-function synthesis and the deconvolution operator used in [Section 9.3](#).

References

- [1] J. Arias de Reyna, *An infinitely differentiable function with compact support: definition and properties*, English translation of *Rev. Real Acad. Cienc. Exact. Fis. Natur. Madrid* **76** (1982), 21–38, arXiv:1702.05442, 2017.

- [2] J. Arias de Reyna, *Arithmetic of the Fabius function*, arXiv:1702.06487, 2017.
- [3] NIST Digital Library of Mathematical Functions, Chapter 17, *q-Hypergeometric and Related Functions*, especially Section 17.2, release 1.2.7 of 15 June 2026.
- [4] J. Fabius, A probabilistic example of a nowhere analytic C^∞ -function, *Zeitschrift für Wahrscheinlichkeitstheorie und Verwandte Gebiete* **5** (1966), 173–174, doi:10.1007/BF00536652.
- [5] G. Gasper and M. Rahman, *Basic Hypergeometric Series*, second edition, Encyclopedia of Mathematics and its Applications 96, Cambridge University Press, 2004.
- [6] J. K. Haugland, *Evaluating the Fabius function*, arXiv:1609.07999, 2016, revised 2020.
- [7] B. Jessen and A. Wintner, Distribution functions and the Riemann zeta function, *Transactions of the American Mathematical Society* **38** (1935), 48–88.
- [8] V. Kac and P. Cheung, *Quantum Calculus*, Universitext, Springer, 2002.
- [9] Z. F. Koçak and G. M. Phillips, B-splines with geometric knot spacings, *BIT Numerical Mathematics* **34** (1994), 388–399, doi:10.1007/BF01935648.
- [10] S. L. Lee and G. M. Phillips, Polynomial interpolation at points of a geometric mesh on a triangle, *Proceedings of the Royal Society of Edinburgh Section A: Mathematics* **108** (1988), 75–87, doi:10.1017/S0308210500026536.
- [11] V. Reshetnikov, *The Fabius Function and Rvachev’s Up-Function: Exact arithmetic, probability, Fourier analysis, and corrected sharp asymptotics*, ProveIt repository, Analysis/FabiusFunction/docs/Fabius_Function_and_Rvachev_Up, repository snapshot inspected 30 August 2026.
- [12] V. Reshetnikov, *Lagrange–Legendre–Rvachev loop report*, ProveIt repository, Analysis/FabiusFunction/docs/semi-formalized-research-frontiers/drafts/representation/s/lagrange_rvachev_loop_report_v3, repository snapshot inspected 30 August 2026.
- [13] I. J. Schoenberg, On polynomial interpolation at the points of a geometric progression, *Proceedings of the Royal Society of Edinburgh Section A: Mathematics* **90** (1981), 195–207, doi:10.1017/S0308210500015456.